

BMCS 2203
BMCS 2074
BMCS 2003
Artificial Intelligence
Tutorial Book

Lecturer/Tutor:

Tutorial 1

1. Who is the father of A.I.? **Describe** the reason why he was recognized so. ()

Answer:

The father of AI is John McCarthy

"making a machine to behave in ways that would be called intelligent if a human were so behaving." by John McCarthy

He created a term of AI (making a machine to behave in ways that would be called intelligent if a human were so behaving.)

Dartmouth Conference: objective of the conference to explore ways to invent a machine that could reason like human

2. Identify **one key event** or major achievement of A.I. development in the year of
- a) 1960-1969 ()
 - b) 1970-1979 ()
 - c) 1980-1989 ()
 - d) 1990-1999 ()
 - e) 2000-2009 ()
 - f) 2010-2019 ()
 - g) 2020-now ()

Answer:

1960–1969:

ELIZA, an early natural language processing program, was developed in 1966 by Joseph Weizenbaum.

ELIZA - chatbot that hold conversation in human

Unimate - first industrial bo

1970–1979:

The development of expert systems like MYCIN in the mid-1970s, which could diagnose bacterial infections.

Intro of the Expert System MYCIN-designed to diagnose blood infection

Development of Wabot-i : first human -sized robot. Simulate human conversation and perform basic tasks like walking

1980–1989:

A.I. entered the commercial world through expert systems, especially in industries like finance and medicine.

Rise of Neural Network (1986): introduced the backpropagation algorithm, leading to advances in machine learning

Development of Polly- behavior-based robotics

Commercial expert system- used in automotive industries to diagnose car engine problems

1990–1999:

In 1997, IBM's Deep Blue defeated world chess champion Garry Kasparov, showing A.I.'s capability in strategic game play.

Development of ERS-110 Robot Pet Dog (1999) - interact with human in a more personal and emotional way

Development of ALICE (1995) - chatbot, contribute to data collection and research in NLP

2000–2009:

Speech recognition technology improved, and IBM's Watson project began. IBM's question answering computer Watson wins first place on popular \$1M prize television quiz show Jeopardy

Roomba - autonomous robotic vacuum cleaner

advancement in recommendation technology - rapid growth and increased computational capabilities. Eg. Netflix is using algorithm to suggest movie to users

2010–2019:

In 2017, Google's AI AlphaGo beats world champion Ke Jie in the complex board game of Go notable for its vast number (2^{170}) of possible position

Siri, Apple virtual assistant, voice activated assistant, interpret and respond to user commands

Launch of Cortana-developed by Microsoft streamline user productivity

2020–now:

The release of ChatGPT and GPT-4 by OpenAI significantly advanced natural language understanding and generation.

ChatGPT-generating human-like text for various application

GitHub Copilot - suggesting code or complete programming task

3. Name an artificial intelligence application that is created by a Malaysian company. Briefly describe its A.I. functions. ()

Answer:

ADAM by **Fusionex** is an A.I. chatbot and virtual assistant. It uses natural language processing to understand customer inquiries and machine learning to improve responses over time, helping businesses automate customer service and support.

AskAILA

- Legal assistant developed by the Malaysian law firm Shang & Co
- Provide accessible legal advice at lower cost

4. Differentiate between the following, then provide an example for each type of system.

- a) Systems that think like humans. ()
- b) Systems that act like humans. ()
- c) Systems that think rationally. ()
- d) Systems that act rationally. ()

Answer:

Systems that think like humans - machine learning

- These systems try to copy how people think or solve problems.
- **Example:** A program that learns how students solve math problems, like how a teacher thinks.
- The cognitive modelling approach
 - Designed to solve problems by thinking, reasoning, and remembering, to mimic the way the human brain works
 - Example: IBM Watson, can understand , process and respond to human language such as answering questions

Systems that act like humans - automation

- These systems behave like a real person.
- **Example:** A robot that talks and moves like a human, such as **Sophia the robot**.
- The Turing Test Approach
 - Perform human's level action or interaction with minimal human assistance like communication in human language without understanding the meaning of action
 - Example: Siri - can understand to natural language, interact with someone in conversation

Systems that think rationally - expert system

- These systems follow rules and logic to think and make decisions, like solving puzzles or doing calculations.
- **Example:** A computer system that helps doctors decide on treatments using rules.

- **Logic & Rule-based system**
 - Making inferences based on rules
 - Reasoning processes are applied to derive new representations of the world
 - Example: MYCIN - diagnosing bacterial infections and recommending antibiotics

Systems that act rationally - adaptive system

- These systems try to do the best or smartest action, even if they don't act like people.
- **Example:** A self-driving car that chooses the safest and fastest way to drive.
- **Agent Approach**
 - Sensitive to environment and react to changes
 - Agent is something that perceives and acts
 - Example: self-driving cars, autonomous vehicles make decisions based on the data from their environment

5. Describe input (conditions), processing layer, output and how the agent react for each of the system below:
- Agent of an automated class scheduling system. ()
 - Agent of an intelligent air-conditioning system. ()
 - Agent of autonomous driving car. ()

Answer:

Agent of an automated class scheduling system:

- **Input:** Course list, lecturer availability, classroom capacity.
- **Processing:** Match courses with available time slots and rooms using scheduling algorithms.
- **Output:** Timetable schedule for classes.
- **Reaction:** Adjust schedule automatically if a conflict or change occurs.

Agent of an intelligent air-conditioning system:

- **Input:** Room temperature, air moisture, number of people in the room.
- **Processing:** Analyze data using fuzzy logic 模糊逻辑 or machine learning.
- **Output:** Adjusted fan speed and temperature.
- **Reaction:** Continuously updates output as input conditions change.

Agent of autonomous driving car:

- **Input:** Road signs, obstacles, GPS data, speed of nearby vehicles.
- **Processing:** Real-time decision-making using AI algorithms (e.g., neural networks).
- **Output:** Steering, braking, accelerating 加速.
- **Reaction:** Reacts to changes in traffic, road conditions, or pedestrian movement to ensure safety.

Sample Answer

Agent of an automated class scheduling system.

Input Layer:

- Course list, Time slots
- Classroom availability
- Student availability

Processing Layer:

- rules for scheduling (e.g., no overlapping class)
- match input data with rules
- handles conflicts

Output Layer:

- generate a complete class schedule

Reaction:

- Agent analyzes input, handles conflicts, and output schedule

Agent of an intelligent air-conditioning system

Input Layer

- Room temperature (sensor)
- Environmental data (outdoor temperature)
- User preferences

Processing Layer

- Determines the required cooling / heating adjustments
- Adapts to user behaviour

Output

- Adjust AC settings (eg: temperature, fan speed)

Reaction

- Agent monitor the room environmental, process user input, adjust settings

Agent of autonomous driving car

Input

- Sensor (camera, gps)
- User input (destination)
- External data (traffic, weather)

Processing

- Perception (object detection, lane detection)
- Decision making (plans the route)

- Control (steering, speed)

Output

- Safe navigation and real time adjustment

Reaction

- Agent processes sensory inputs and external data, make decision, to control the car, ensure safe

Tutorial 2

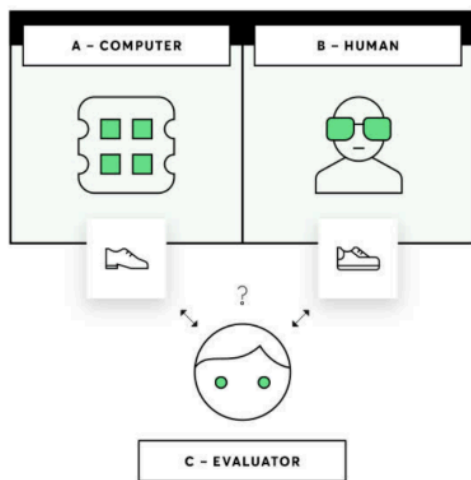
Instead of asking, "Can machines think?", Alan Turing said we should ask, "Can machines pass a behavior test for intelligence?". Turing predicted that by the year 2000, a computer could be programmed to have a conversation with a human interrogator for five minutes and would have a 30% chance of deceiving the interrogator that it was a human. (Negnevitsky, 2002).

1. Explain Turing Test. Illustrate your answer with an appropriate diagram. ()

Answer:

Turing Test

- A method proposed by Alan Turing in 1950 to assess whether a machine can exhibit intelligent behavior equivalent to, or indistinguishable from, that of a human.
- 图灵测试是艾伦·图灵于 1950 年提出的一种方法, 用于评估机器是否能够表现出与人类相当或无法区分的智能行为。



The "standard interpretation" of the Turing test:

An interrogator, who is a human, is given the task of trying to determine which player – A or B – is a computer and which is a human.

If the machine is able to deceive the interrogator, then the machine passes the Turing test and it is considered to be intelligent.

Turing test: There are three participants

1. A human judge (interrogator): Conduct test and interacts with both the human and the machine
 2. A machine: being tested for its ability to exhibit intelligent behavior
 3. A human: Serves as the real-human benchmark for comparison
- All conversation happens via a text-based interface.
 - The judge asks questions and compares the two sets of answers.
 - If the judge can't consistently tell which participant is the machine, the machine "passes" the Turing Test and is considered intelligent.

2. Criticize Turing's criteria for judging a computer's intelligence. ()

Answer:

1. Focus on Imitation, Not Understanding

- The Turing Test checks if a computer can *act* like a human in a conversation. But acting like a human doesn't mean it really *understands* anything. It could just be using smart tricks or memorized answers.

2. Human-Like ≠ Intelligent:

- The test thinks that being like a human means being intelligent. But a machine can be intelligent in its own way (like playing chess or solving complex math), even if it doesn't talk like a human. So, the test is too limited.

Lacks Contextual Understanding

- Focuses on mimicking conversation, not true understanding
- Machine may give convincing replies without comprehending meaning

Limited Scope (Only Language-Based)

- No visual or physical interaction
- Language alone does not prove full human level intelligence

Ignore Non Verbal Intelligence

- Overlooks skills like problem-solving, creativity, and emotion
- Only tests linguistic ability, not overall intelligence

3. Suggest how this test could be used (or modified) to assess other kind of artificial intelligence besides a chatbot. Provide an example to elaborate your answer. ()

Answer:

The Turing Test can be changed to test other types of AI, not just chatbots. For example, we can test a **self-driving car AI**. A human and the AI can both drive in the same driving simulator. If a judge cannot tell which driver is human and which one is AI, then the AI has passed the test. This checks if the AI can drive like a human, not just talk like one.

Without Modification: CAPTCHA (Reverse Turing Test)

- Inspired by Turing Test, but works in reverse Purpose:
- Turing Test: Checks if a machine can act human
- CAPTCHA: Checks if a user is human or a bot

Examples:

- Text-Based CAPTCHA: User types distorted text, Easy for humans, hard for bots
- Image-Based CAPTCHA: User selects images with specific objects (e.g., cars), Tests visual recognition ability

Text-Based: https://static.zenrows.com/content/medium_text_captcha_demo_1908e3a0f6.jpg

Image-Based:

<https://www.researchgate.net/publication/349180355/figure/fig2/AS:1132272803692546@1646966281751/>

Example-of-the-image-based-CAPTCHA-a-Click-image-based-CAPTCHA-b-Slide-image-based.png

With modification

1. Task Oriented Evaluation
 - Test AI on domain specific tasks (eg: decision making, creativity)
 - Eg: AI recommends medical treatments or financial investments
2. Contextual Understanding
 - AI must respond based on context or user specific info
 - Eg: AI gives medical advice tailored to a patient's history

4. The Loebner Prize is an annual competition in artificial intelligence that awards those computer programs considered by the judges to be the most human-like, using the format of a standard Turing Test. The conversation scope between the programs and the judges has been unrestricted since 1995, and the duration of the conversation has been increased from 5 minutes to 25 minutes since 2010 (<http://www.loebner.net/>). ()
- Discuss **TWO (2)** reasons why the Turing Test is considered **not effective enough** in assessing machine intelligence.
 - Discuss **TWO (2)** challenges to build a computer program that can win the Grand Loebner Prize, in which judges totally cannot distinguish it from a real human.

Answer:

a)

1. Only Test Conversation

- The test only checks if a machine can talk like a human. It doesn't check if the machine can learn, solve problems, or understand emotions.

2. Depends too much on judge

- The result can change based on the judge's skills or mood. Some judges may be easy to trick, others may not.

1. Imitates, Not Understand

- Focuses on mimicking human response, not true comprehension
- The test fails to distinguish between machines that truly understand and those that only simulate understanding

2. No Objective Measurement

- Relies on human judgement, which is subjective
- Lacks clear standards or scoring system

b)

1. Understanding complex human language:

- Humans use jokes, emotions, and sarcasm. It is very hard for a computer to understand and reply to these like a human.

2. Keeping the conversation natural for 25 minutes:

- The AI must stay interesting, give good answers, and not repeat itself for a long time, which is very difficult.

1. Natural language and context

- Must understand idioms, sarcasm, humor, and cultural references
- Needs deep NLP and consistent context awareness

2. Emotional and empathetic response

- Hard to detect and respond with the real empathy

- Requires emotionally intelligent, human like replies

3. Language complexity

- Human language constantly evolves (eg: new slang, terms)
- Different languages = different rules and expressions

5. The Chinese room argument by John Searle is one of the best known and widely credited criticism of the Turing Test. Explain John Searle's Chinese room concept. ()

Answer:

John Searle said that even if a computer gives correct answers, it doesn't mean it really understands them.

He gave an example:

Imagine a person who doesn't know Chinese sits in a room. People pass Chinese sentences into the room. The person uses a book (with rules) to match the correct Chinese answers and send them out. To outsiders, it looks like the person understands Chinese—but inside, the person is just following rules without understanding.

This shows that a computer might give smart answers, but it doesn't really "understand"—it just follows code.

- A computer running a program may behave intelligently but does not truly understand
- Searle imagines himself in a room, following rules to reply to Chinese characters he doesn't understand
- Outsiders think he understands Chinese, but he is just manipulating symbols
- Similarly, a computer may pass the Turing test but still lack real understanding or consciousness

Conclusion: Running a program $\neq$ having a mind or true understanding

6. Try to chat with the following chatbots within a few minutes. Then discuss what are the characteristics/behaviors a chatbot should have in order to deceive any human. ()
- Mitsuku**, the 5-time Loebner Prize winner - <https://www.pandorabots.com/mitsuku/>
 - Eliza**, the first chatbot - <https://web.njit.edu/~ronkowitz/eliza.html> (not the original Eliza website)

Answer:

- Good memory and context understanding:**
A smart chatbot should remember things you said earlier and give answers that make sense.
- Emotion and personality:**
It should reply with feelings (happy, sad, funny) like a real person, not just give robotic answers.
- Correct grammar and natural language:**
The chatbot should use proper grammar and talk like humans do—not sound too perfect or too simple.
- Handle unexpected questions:**
If you ask a strange question, the chatbot should still try to give a smart answer, not get confused or repeat the same thing.

Mitsuku - Key Features

Natural Language Understanding

- Understands context and complex sentences

Contextual Memory

- Remembers past conversations for continuity

Continuous Learning

- Improves over time through user interactions

Mitsuku's advanced language skills, memory, emotional responses, and learning ability make it a highly engaging and intelligent chatbot.

Eliza - Key Features

Rule-Based System

- Uses predefined patterns and keyword matching

Limited Scope

- Focused on basic psychotherapy topics only

Text-Based Interface

- Interacts through simple text, no visuals or media

Eliza uses a rule-based, text-only approach with limited understanding and context, but played a key role in early chatbot development.

Tutorial 3

1. Discuss the advantages and disadvantages of **breadth-first search** and **depth-first search**. ()

Answer:

Breadth-First Search (BFS):

- **Advantages:**

- Guaranteed to find the shortest path (optimal solution) if one exists.
- Complete — will find a solution if one exists.
- Guarantees the shortest path - find minimum number of steps
- Faster when the goal is near the root - goal is located at high level

- **Disadvantages:**

- Consumes a lot of memory (space complexity: $O(b^d)$, where b is the branching factor and d is the depth).
- Slower in deep trees due to expanding all nodes at a level before going deeper.
- High memory usage - store all branches
- Slow with large branching factors - longer search time

Depth-First Search (DFS):

- **Advantages:**

- Requires less memory than BFS (space complexity: $O(b \cdot d)$).
- Can find a solution quickly if the solution is deep in the tree.
- Low memory usage - only stores the nodes in the search path
- Usually the search time is faster

- **Disadvantages:**

- Not complete in infinite search spaces.
- Not optimal — may not find the shortest path.
- Does not guarantee the shortest path
- Risk of infinite loops
- Can still consume memory in deep searches - if goal is very deep down

2. **Figure 1** shows a puzzle problem that requires rearrangement of the tiles to transform the order from start to goal state. One is only permitted to slide the empty tile follow the sequence of **left, right, up or down**.

Remark: Avoid repeated state

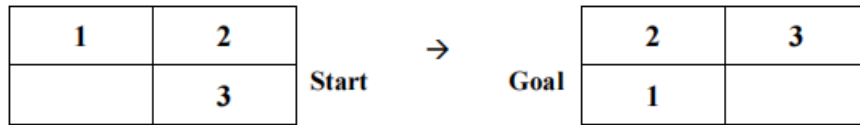


Figure 1: The Puzzle Problem

- Provide the goal formulation of the puzzle problem above. ()
- Formulate the puzzle problem above by specifying the initial state, successor functions, goal test, step cost, and path cost. ()
- Perform **breadth-first search** and **depth-first search** on the puzzle problem above. Draw the resulting search trees for both. ()

Answer:

- a) **Goal formulation:** to reach the goal as shown in Figure 1

2 3

1

Optimal Solution: find the shortest path to reach the goal

Abstraction : time

- b) **Initial State:** start state

1 2

3

Successor Function: Generate all possible legal moves of the empty space (left, right, up, down) that don't go outside the grid

Goal Test: State matches the goal configuration.

Step Cost: 1

Path Cost: Total number of moves made to reach the goal

Initial State: start state

Successor function: All possible movements to successor state from current state

Goal Test: to determine whether a given state is a goal state, otherwise continue to search

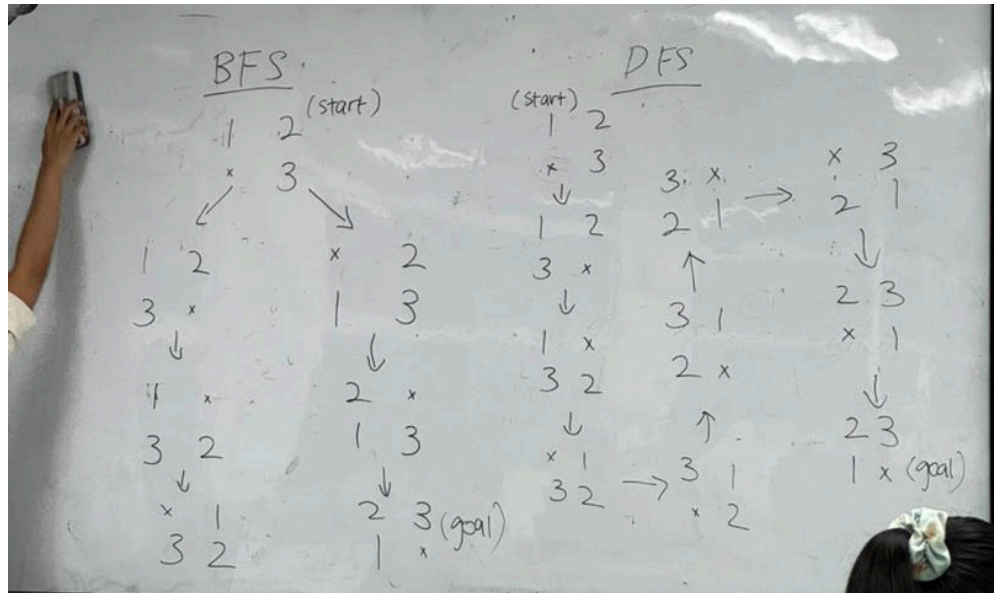
Step Cost: the distance between one node to the other

Path Cost: the total step cost between start state and the goal state along a path

c) Search Trees

sequence of left, right, up or down.

Is the x moving left, right, up, down



3. **Figure 2** shows a puzzle problem that requires rearrangement of the tiles to transform the order from start to goal state. One is only permitted to slide the empty tile follow the sequence of **down, right, up or left**
Remark: Avoid repeated state

1	2	3	→	1	2	3
	5	6		4	5	6
4	7	8		7	8	
Start			Goal			

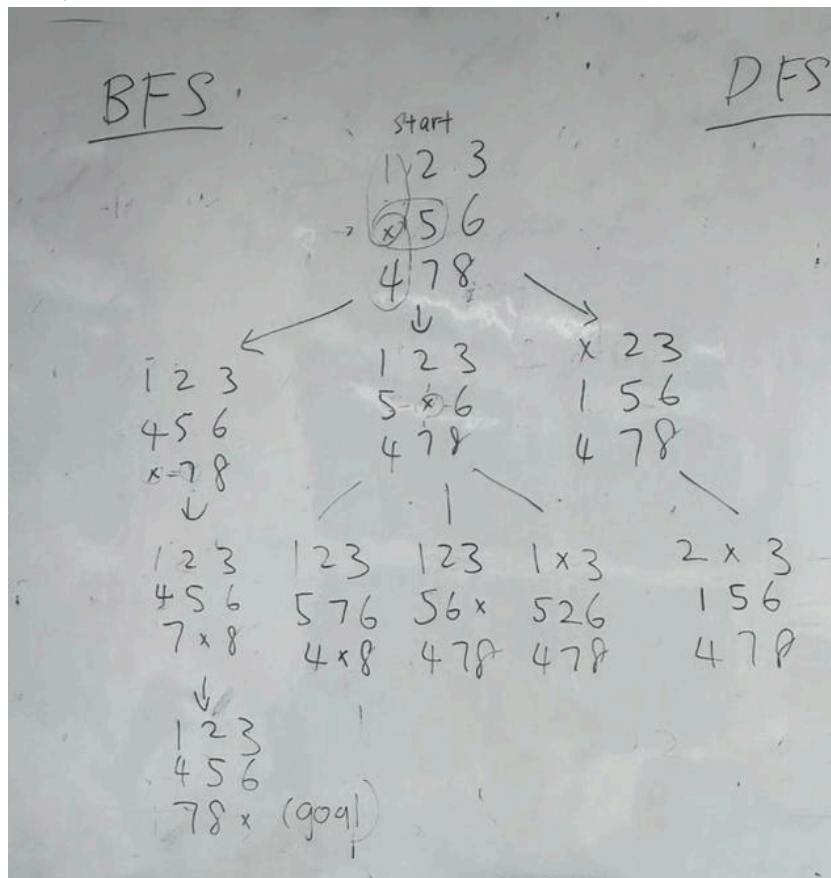
Figure 2: The 8-Puzzle Problem

- a) Perform **breadth-first search** and **depth-first search** on the 8-puzzle problem above. Draw the resulting search trees for both. ()
- b) Evaluate the efficiency of **breadth-first search** and **depth-first search** in terms of completeness, optimality, time efficiency and space efficiency in solving the problem above. ()

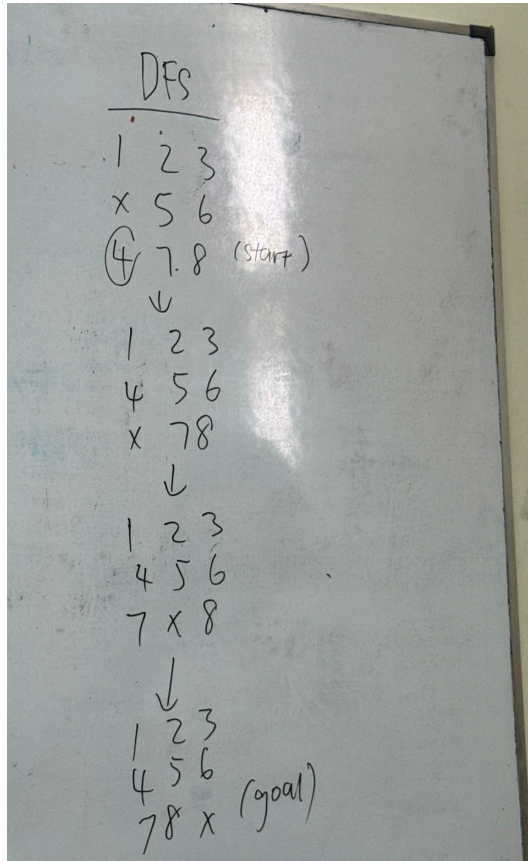
Answer:

down → right → up → left

a) BFS



DFS



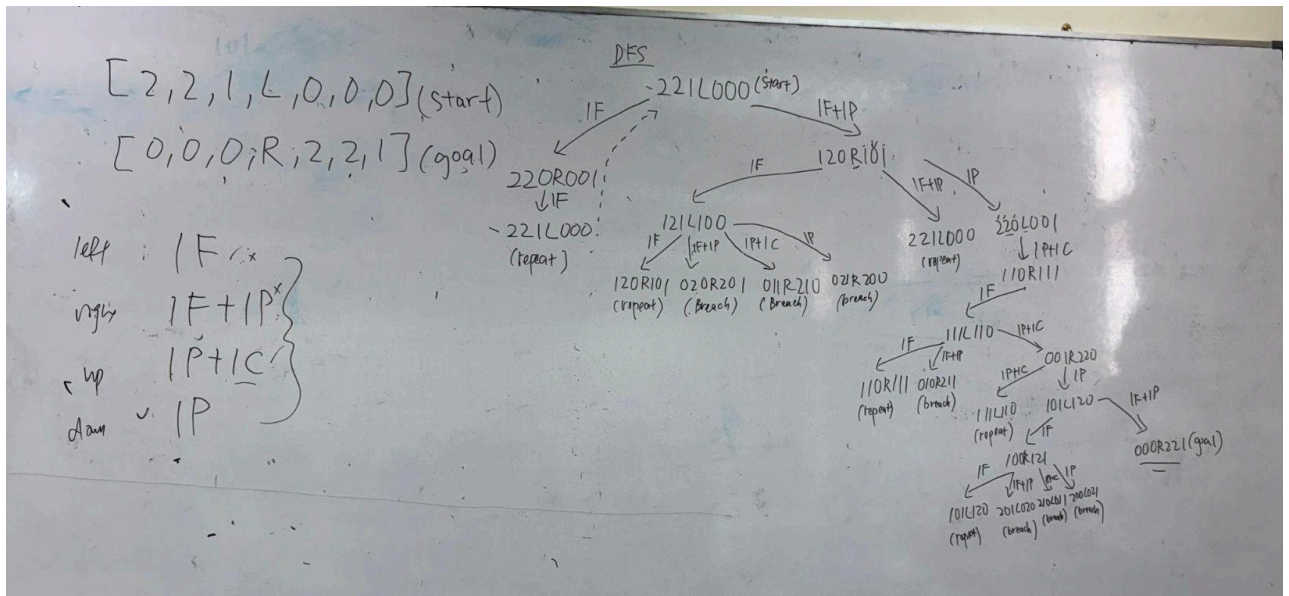
b) Efficiency of breadth-first search and depth-first search

Efficiency	Breadth-first search	Depth-first search
Completeness	Yes	Yes
Optimality	Yes	Yes
Time complexity	Consume more time	Consume less time
Space complexity	Consume more memory	Consume less memory

4. In the family river-crossing problem, two parents are with their two children - a son and a daughter - came to a wide river. The only way to get to the other side was to ask a fisherman if he could lend them his boat. However, the boat could carry only two persons. For safety reason, no child should be left alone without the supervision of at least one parent. The family must get to the other side and finally returns the boat to the fisherman, assuming only the fisherman and the two parents know how to row the boat. Based on this family river-crossing problem, answer the following questions.

Remark: Avoid repeated state

- Suggest a simple representation of the initial state. You must briefly explain the representation. ()
- Describe the goal formulation and problem formulation: initial state, successor function, step cost, path cost and goal test. ()
- Draw the **depth-first search tree** to show how all the states are being traversed. ()



5. **Figure 3** below shows a directed graph. Assume that the traversal would start from **Vertex 0** to **Vertex 7**. All vertices to be visited in **ascending order** (i.e. from smaller number to bigger number).

Remark: Avoid repeated state

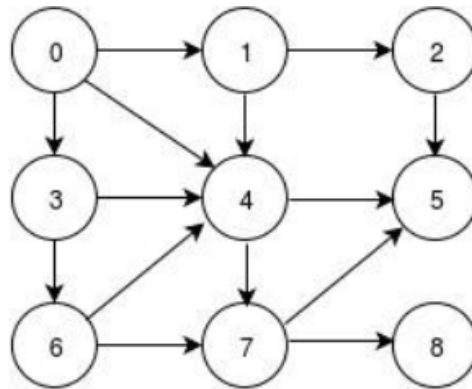
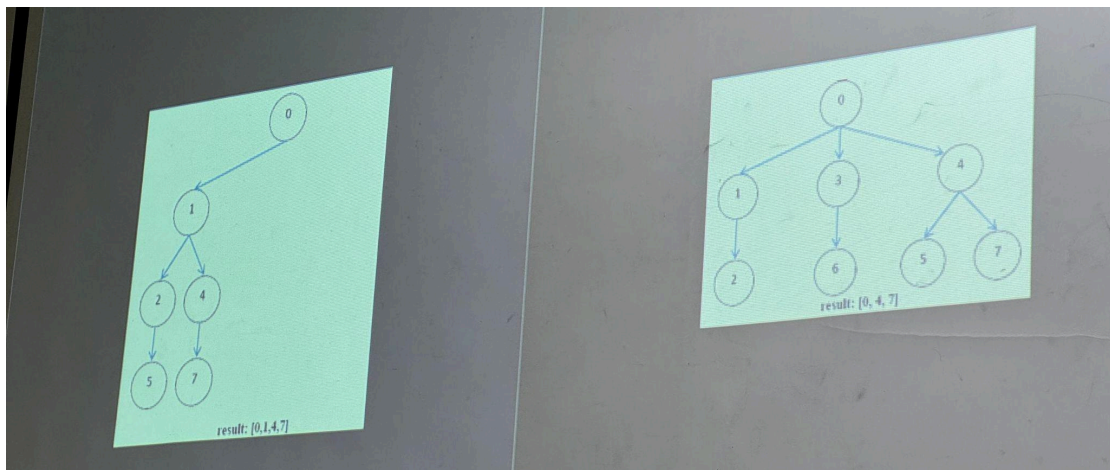


Figure 3: The Directed Graph

- Perform a **depth-first search** on the directed graph to traverse from Vertex 0 to Vertex 7. Draw the resulting search tree and list the returned path. ()
- Perform a **breadth-first search** on the directed graph to traverse from Vertex 0 to Vertex 7. Draw the resulting search tree and list the returned path. Avoid repeated state. ()
- Evaluate the efficiency of **breadth-first search** and **depth-first search** in terms of completeness, optimality, time efficiency and space efficiency in solving the problem above. ()

DFS

BFS



c) Efficiency of breadth-first search and depth-first search

Efficiency	Breadth-first search	Depth-first search
Completeness	Yes	Yes
Optimality	Yes	Yes
Time complexity	Consume more time	Consume less time
Space complexity	Consume more memory	Consume less memory

Tutorial 4

1. **Figure 1.1** below shows a Block World Problem. A robot will move the blocks one by one from initial state S1 to reach the goal state S19.

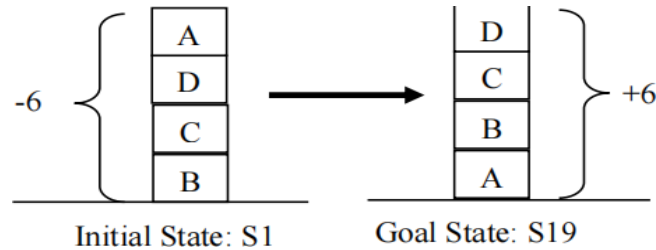


Figure 1.1

Figure 1.2 shows the state space of the Block World Problem and the heuristic costs for each state are shown in parentheses next to their respective nodes.

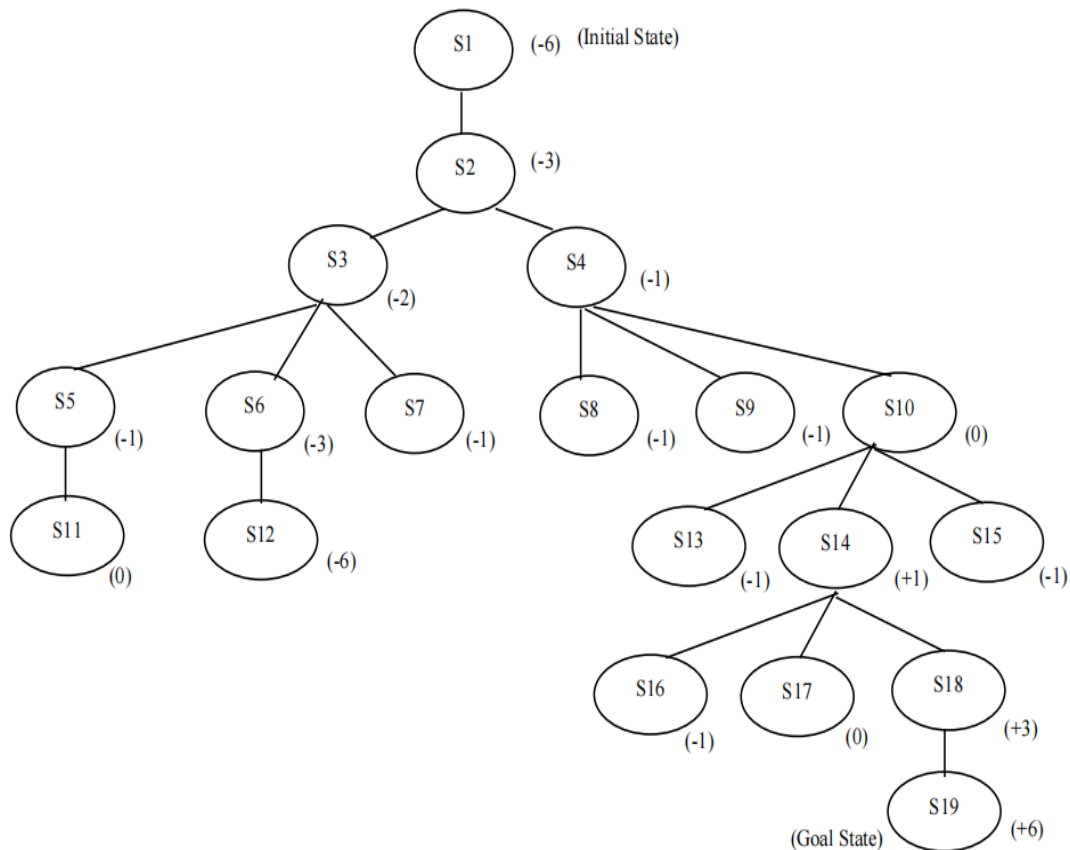


Figure 1.2

- Explain step cost used in problem formulation. Specify the value of the step cost for the problem above. ()
- Hill climbing search is unable to guarantee completeness and optimality as it may be trapped into local maximum. ()
 - Explain local maximum.
 - Discuss why hill climbing search always leads to a local maximum.

Use simple hill-climbing and steepest-ascent hill-climbing to search for the best path from S1 to S19 on the state space shown in **Figure 1.2**. Then for each search technique, draw the resulting search tree that shows the visited nodes. Show that hill-climbing technique can be trapped into a local maximum. (**Remark: Avoid repeated state**)

- c) A search technique can be evaluated based on four criteria: completeness, optimality, time complexity and space complexity. Evaluate the efficiency of breadth-first search and steepest-ascent hill-climbing. Conclude which technique is better to solve the Block World Problem mentioned in **Figure 1.1**.

(Remarks: you are required to perform **breadth-first search before comparing**) ()

a) - step cost means that the cost of doing one action to move from one state to another

- In the block world problem, all actions have the same constant cost
- However, in path-finding problems, stepcosts often vary based on distance (km) or time (seconds) between locations.
- thus , most probably every step cost will g=have a different numeric value
- The value of step cost is 1 (the robot moves one block at time, each move is treated as one steps)

b)

(i) Local Maximum

- A local maximum is a state in the search space that has a higher heuristic value than all its neighboring states, but is not the globally optimal solution. The algorithm gets "stuck" at this point because it appears to be the best option locally, even though better solutions may exist elsewhere in the search space.

(ii) Hill climbing always moves to the neighboring state with the highest heuristic value. If it reaches a state where all neighboring states have lower values (a local maximum), it cannot progress further toward the global maximum. This greedy approach lacks mechanisms to escape local optima.

i) Local Maximum

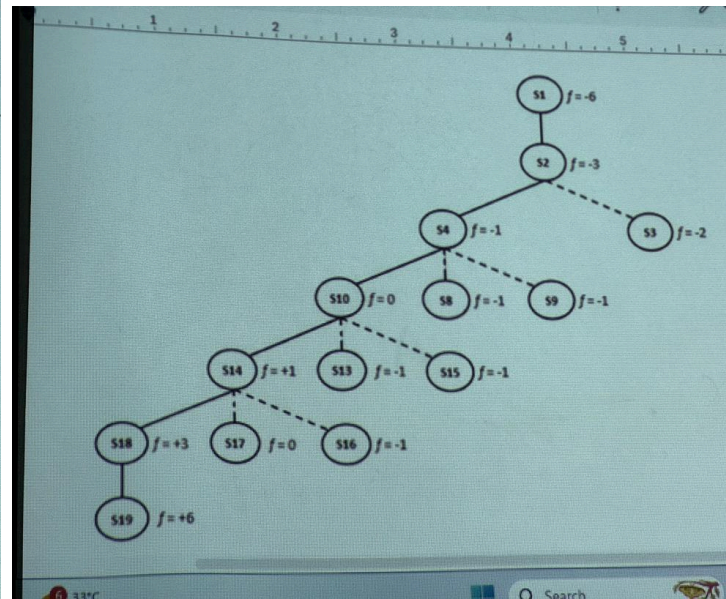
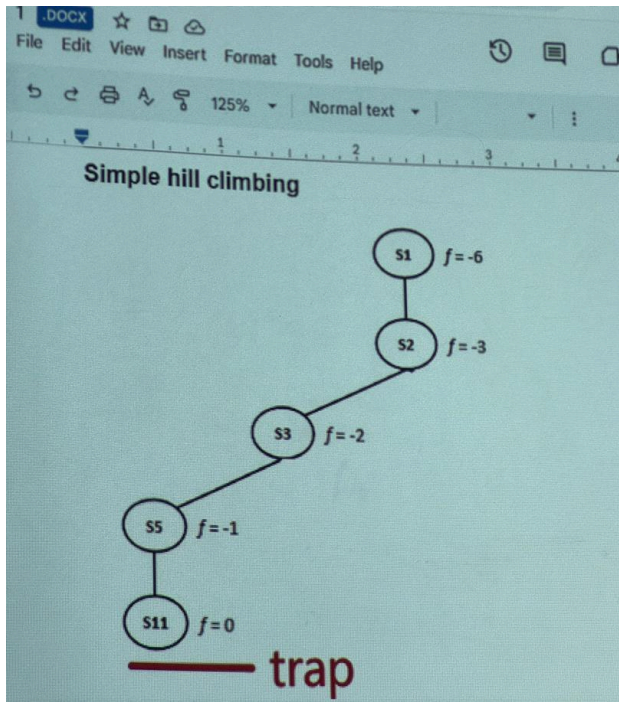
- Refers to states with no better child states or successors
- The presence of local maximum makes a search fails to find a solution (eg 8 queen problem) or an optimal solution (eg travelling salesman problem)
- foothill , lata and ridge are 3 types of local maximum

ii) - the nature of its uphill moves that keep looking for a better successor

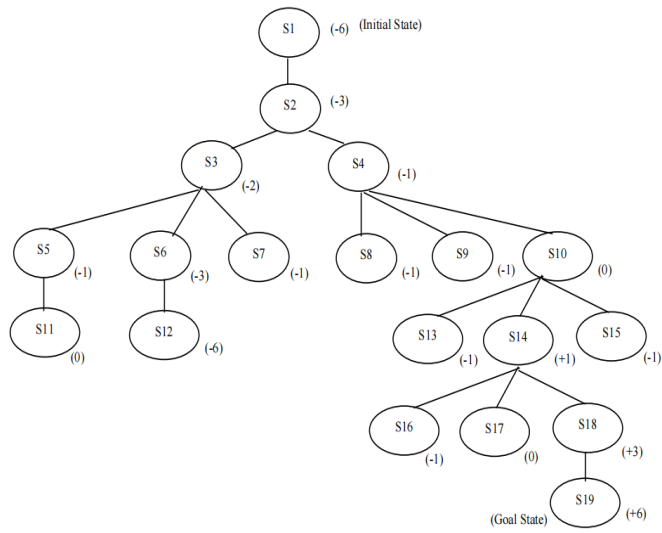
- Unable to backtrack to any least promising unexplored path
- Increase its tendency to reach and end at a peak (or local maximum) where no neighbour has a higher value

c)

Criteria	Breadth-First Search	Steepest-Ascent Hill Climbing
Completeness	Yes (guaranteed to find solution if exists)	No (may get stuck in local maxima)
Optimality	Yes (finds shortest path)	No (may not find optimal solution)
Time Complexity	$O(b^d)$ - high for deep solutions	$O(b)$ per iteration - generally faster
Space Complexity	$O(b^d)$ - stores all nodes at current depth	$O(b)$ - only stores current path



Criteria	Breadth-First Search	Steepest-Ascent Hill Climbing
Completeness	Complete	Complete
Optimality	Optimal (if compared number of station	Optimal
Time Complexity	Has a higher time consumption because it explores all paths by expanding all successors	Has a lower time consumption because it explores only the most promising path by expanding only the best successor
Space Complexity	Has a higher memory consumption because it keeps track of all expanded nodes	Has a lower memory consumption because it doesn't keep track of all expanded nodes



BFS diagram

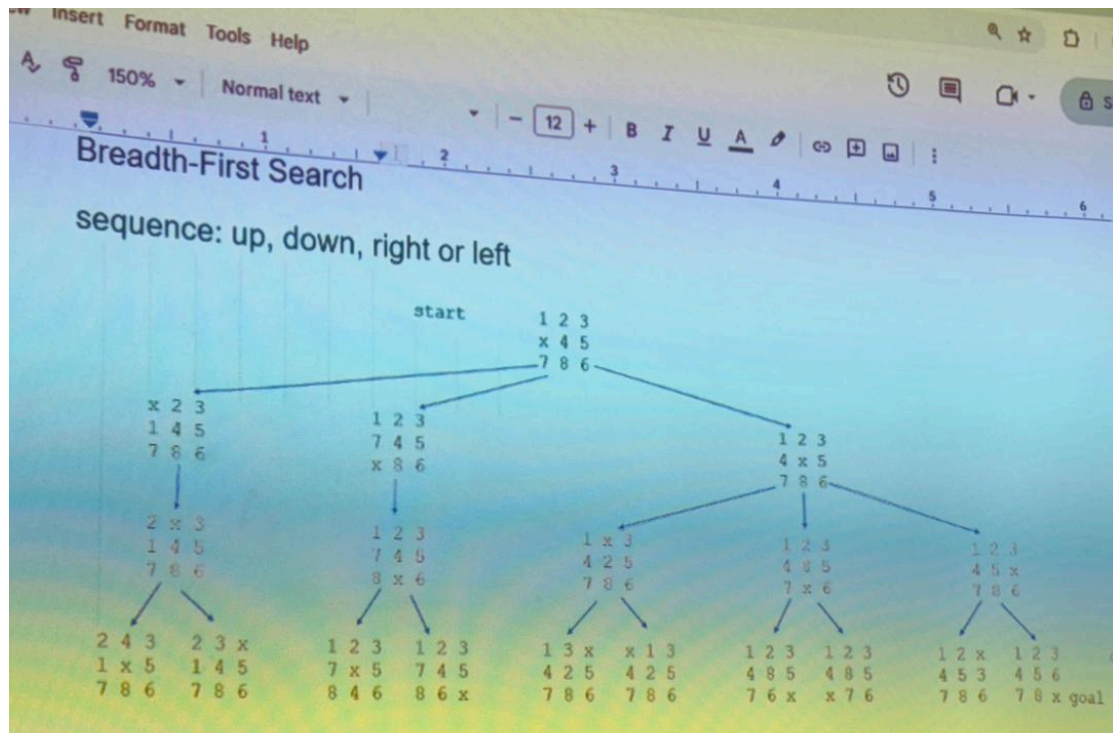
2. **Figure 2** below shows an 8-puzzle problem, which requires rearrangement of the tiles to transform the order from start state to goal state. One is only permitted to slide the empty tile in sequence of **up, down, right or left. (Remark: Avoid repeated state)**

1	2	3	→	1	2	3
	4	5		4	5	6
7	8	6		7	8	
Start				Goal		

Figure 2: The 8-Puzzle Problem

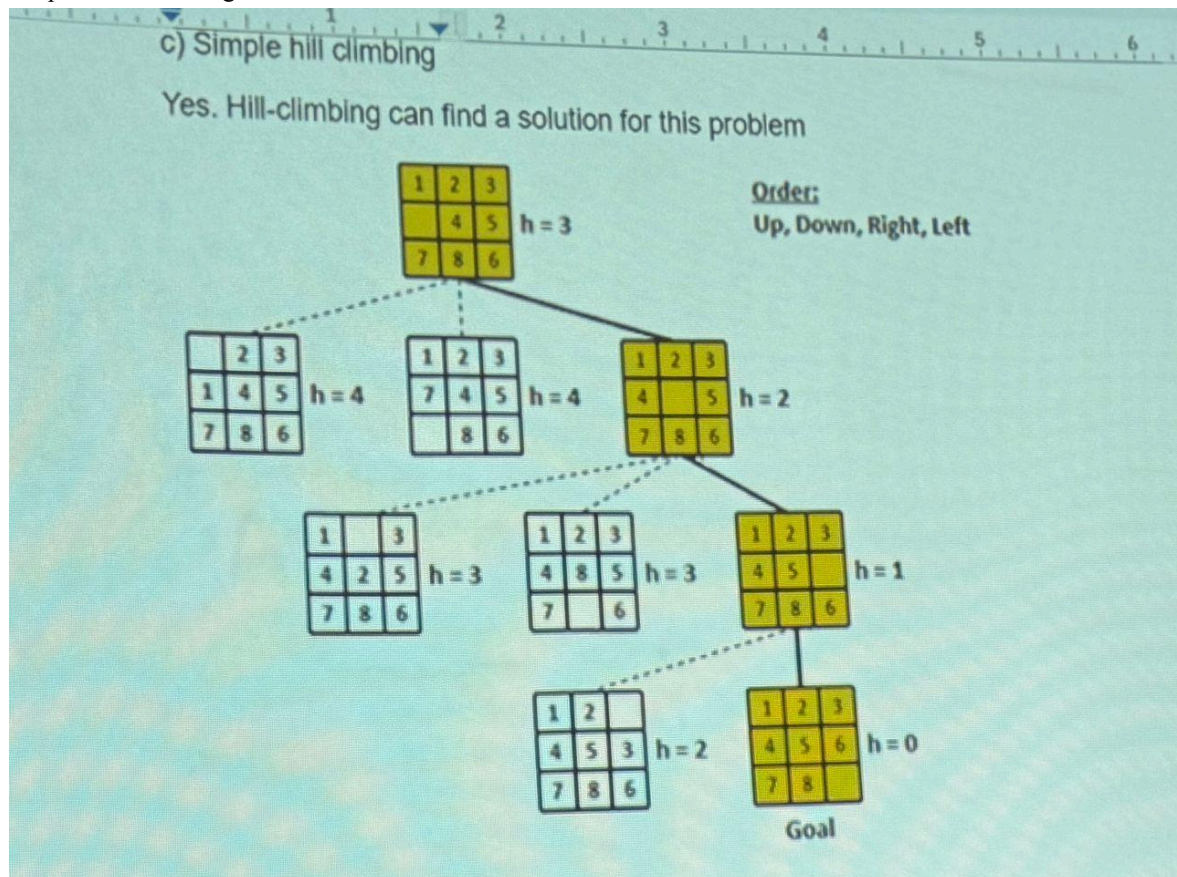
- Suggest a heuristic function to produce a heuristic cost for a state. Demonstrate how such heuristic cost can be computed on the **start state**. Then perform best-first search.
- Evaluate the efficiency of **breadth-first search** and **best-first search** in terms of completeness, optimality, time efficiency and space efficiency in solving the problem above. (Remarks: you are required to perform **breadth-first search before comparing**)
- Can **hill-climbing** find a solution for this problem? Draw a resulting search tree to support your answer.

Heuristic	Description
Hamming distance	The total number of misplaced tiles $f = 0 + 0 + 0 + 1 + 1 + 1 + 0 + 0 = 3$
Best - Fit Search Sequence: up, down, right, left $1+0+0+1+1+1+0+0 = 4$ $0+0+0+1+1+1+1+0 = 4$ $0+0+0+0+1+1+0+0 = 2$ goal:	



Criteria	Breadth-First Search	Steepest-Ascent Hill Climbing
Completeness	Complete	Complete
Optimality	Optimal (if compared number of station	Optimal
Time Complexity	Has a higher time consumption because it explores all paths by expanding all successors	Has a lower time consumption because it explores only the most promising path by expanding only the best successor
Space Complexity	Has a higher memory consumption because it keeps track of all expanded nodes	Has a lower memory consumption because it doesn't keep track of all expanded nodes

Simple Hill Climbing



3. The following graph in Figure 1.1 shows all the nodes in a telecommunication network. The distance (in km) from one node to another is shown on the arc. ()

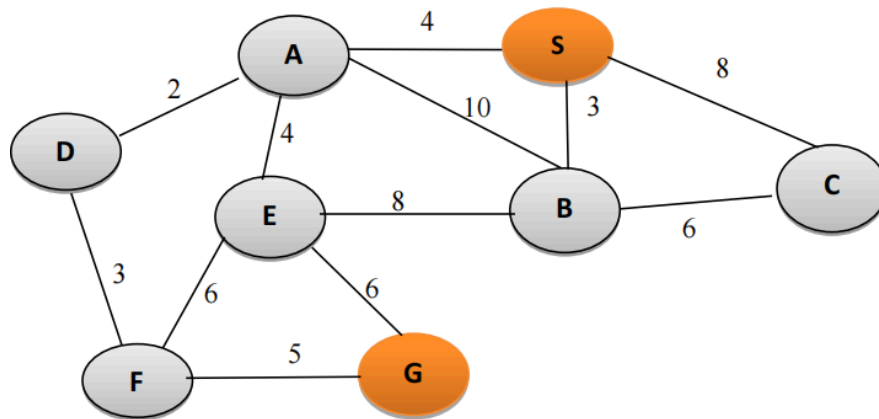


Figure 3: A search graph of a new LRT network

The Euclidean distance (in km), which is used as the heuristic cost (h) for different nodes, is provided in Table 1 below.

S	A	B	C	D	E	F	G
14	10	13	8	8	11	5	0

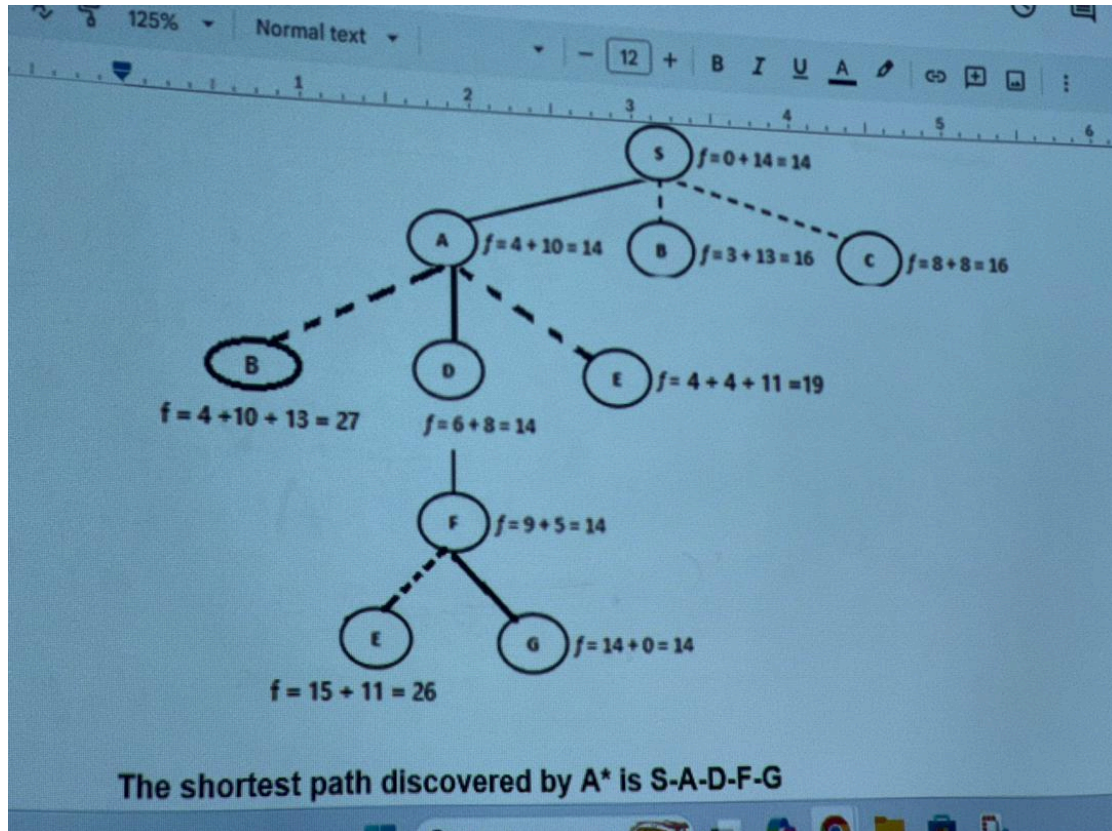
- Assume that some data are to be sent from node S to node G using the shortest route. Describe the goal formulation and problem formulation (initial state, successor function, goal test, step cost and path cost).
- Show the resulting search tree of A* search to find the shortest path from S to G. State the shortest path. (**Remark: Avoid repeated state**)
- Evaluate the efficiency of A* search in solving the path-finding problem above

a)

Goal	Node G
Optimal Solution	The shortest path used to send data from node S to node G.
Abstraction	Ignore the chance that some data might be lost during transmission.

Initial State	Node S
Successor function	The possible actions available to the agent that will change the state from the current state. Eg all possible movement available to the agent
Goal test	A test to check if a node is node G.
Step cost	The cost of transforming a state to another state E.g. The step cost of transforming S to A is 4.
Path cost	The total step costs along a path. E.g, the path cost for S.A.D is $4 + 2 = 6$

b)



c)

	A *
Completeness	Complete
Optimality	Optimal
Time complexity	Time efficiency is high because it explores only the most promising path by expanding only the best successor
Space complexity	Space efficiency is high because it keeps track only expanded nodes.

4. Consider 2 heuristic h_1 and h_2 of A* for the puzzle problem are defined as:

$h_1(n)$ = number of misplaced tiles

$h_2(n)$ = total Manhattan distance

3	1
2	

Start State

Goal State

1	2
3	

- a) Illustrate the **state space** of the puzzle to reach the goal state based on:
- ☐ $h_1(n)$
 - ☐ $h_2(n)$
- b) Show the resulting search trees of A* search to find the shortest path using the heuristic functions of:
- ☐ $h_1(n)$
 - ☐ $h_2(n)$

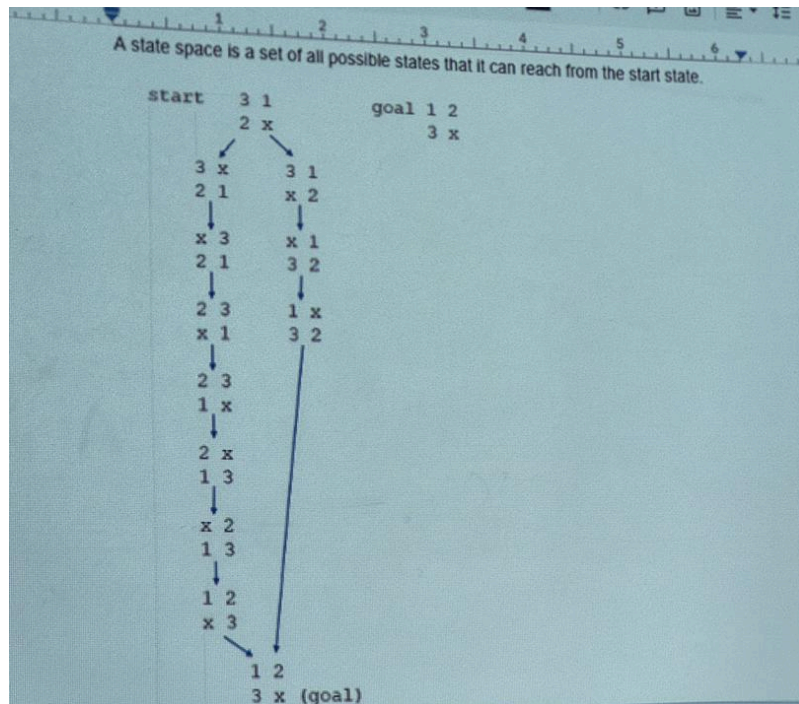
(Remark: Avoid repeated state)

You must clearly show the function cost, given that: $f(n) = h(n) + g(n)$, where $g(n)$ is the path cost.

a)
A* search

3	1
2	

1	2
3	



b)

A* search

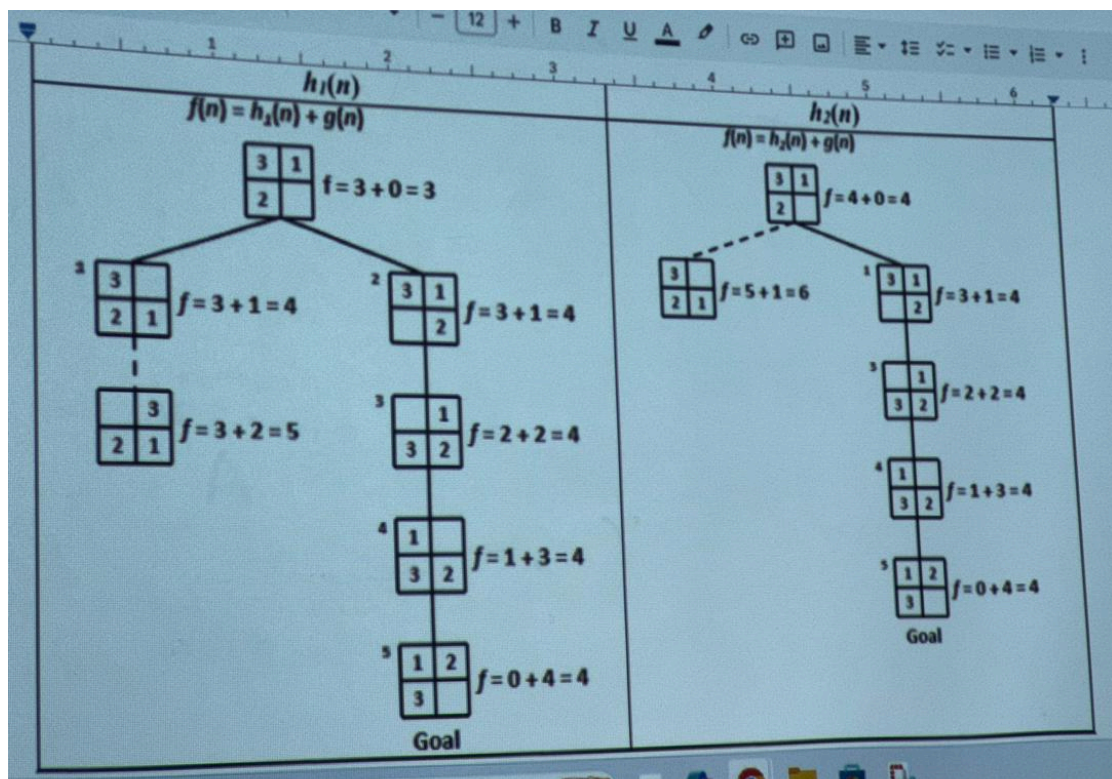
3	1
2	

1	2
3	

$h_1(n)$ = number of misplaced tiles

$h_2(n)$ = total Manhattan distance ()

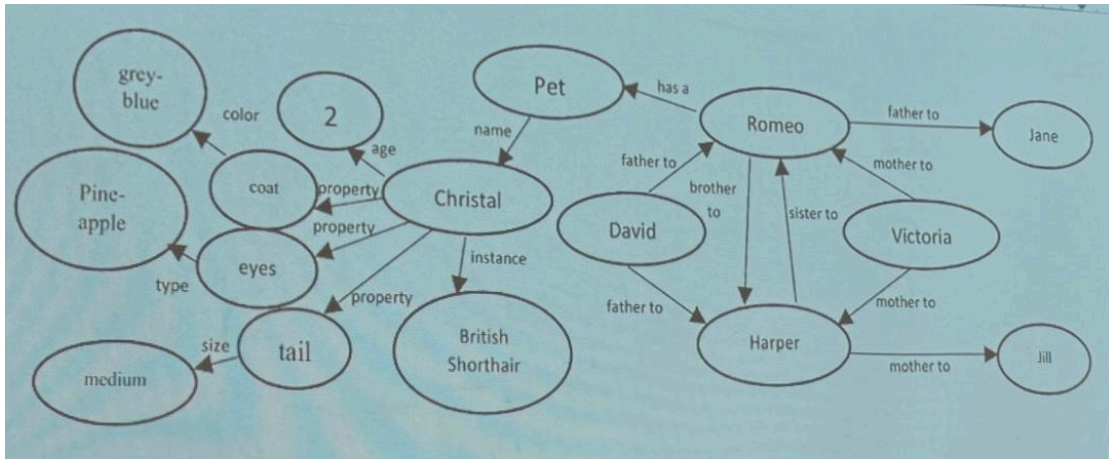
$g(n)$ is the number of moves taken from the start state to reach the current state.



Tutorial 5

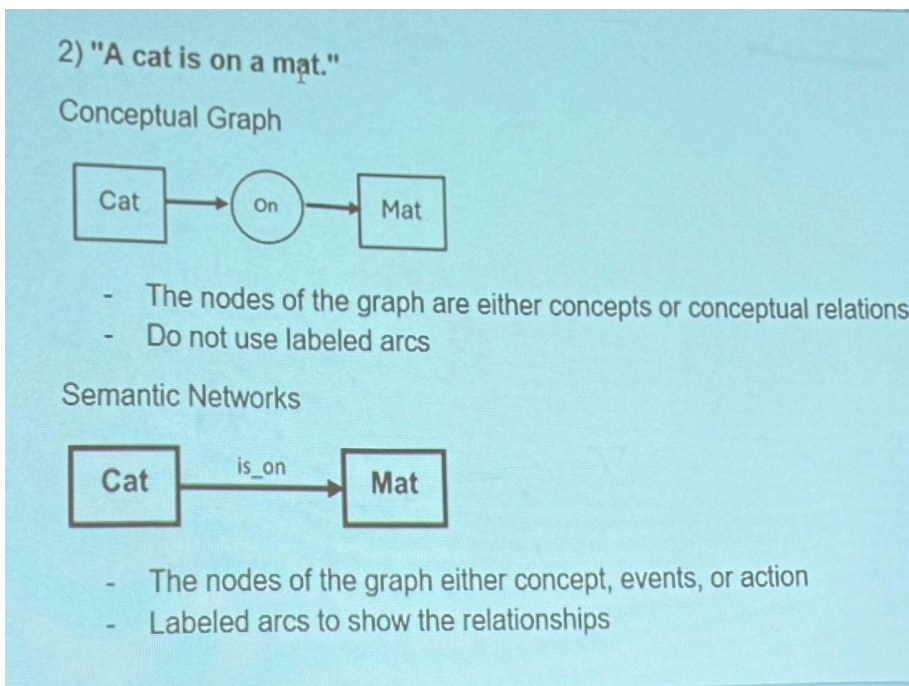
1. A semantic network is a knowledge base that represents semantic relations between concepts in a network. Draw a semantic network for the following facts and relations. ()

"David and Victoria are the parents of Romeo and Harper. Romeo and Harper are brother and sister. Jane is the child of Romeo, and Jill is the child of Harper. Romeo has a pet called 'Christal,' a British Shorthair. Christal is 2 years old now and has a grey-blue coat, pineapple eyes, and a medium-sized tail."



2. How are semantic networks different from Conceptual Graph? Explain your answer with the help of ONE (1) diagram for each of the methods based on the following statement: ()

"A cat is on a mat."



In exam use sentences

3. The following statements are given.

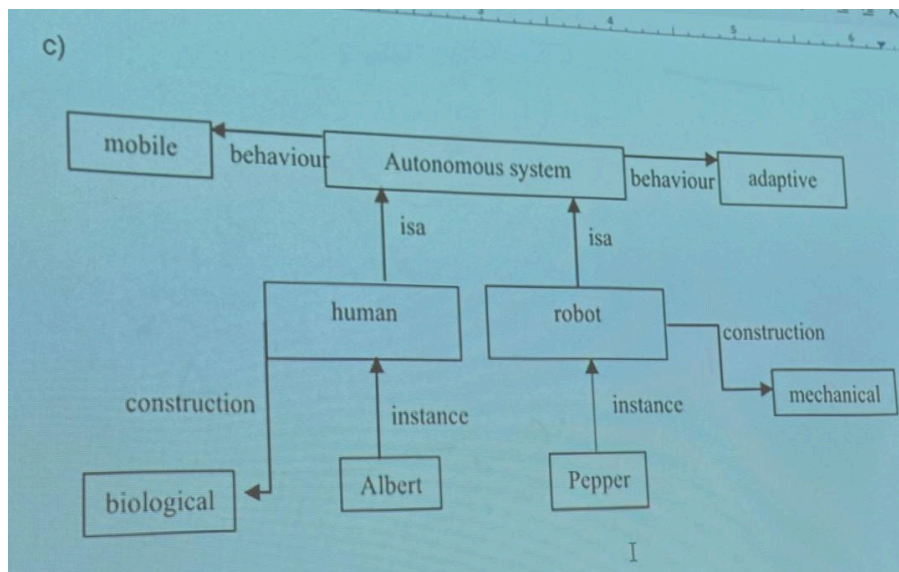
Albert is a human.
Pepper is a robot.
The construction of human is biological, while robot is mechanical.
Both human and robot are autonomous system.
The behaviours of autonomous system are mobile and adaptive.

- Between semantic networks and frames, justify which knowledge representation tool you would use to represent the information above. ()
- Discuss **ONE (1)** limitation of the knowledge representation tool that is selected in Question 3(a) above. ()
- Illustrate the representation of the given statements above using the knowledge representation tool that you selected in Question 3(a) above. ()

Two answer are acceptable

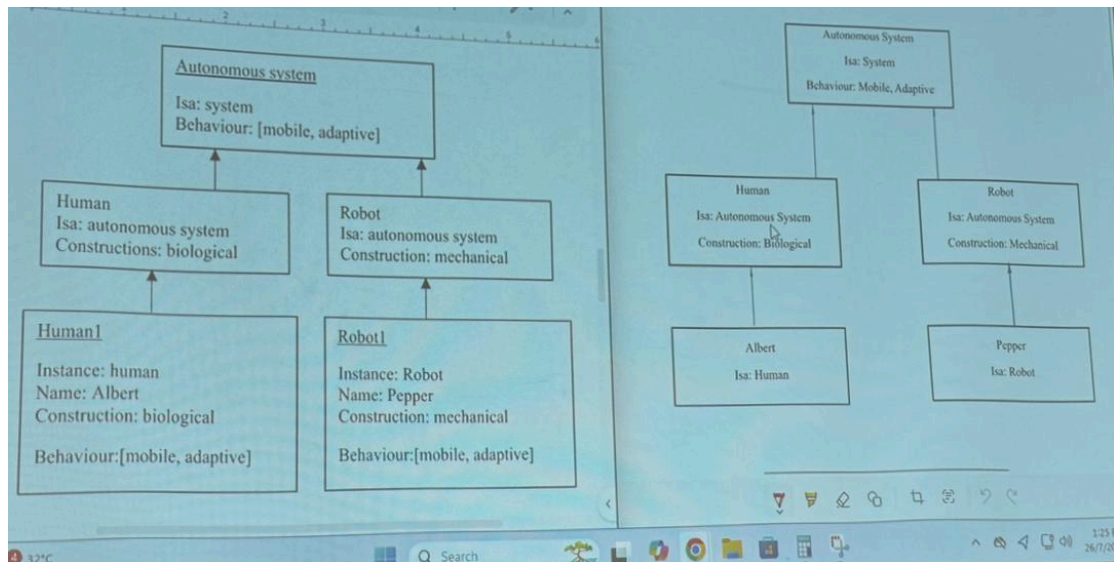
Semantic Network

- Semantic network as it is more suitable to describe the relationship between objects and overall concepts
- Limitation:
 - lack of standardization, no standards about node and arc values
 - difficult to view the relationship between a rule and another.



Frame

- a) Frames as frames are suitable to show the details of objects and their relationships
- b) Limitation:
 - Only able to represent stereotypes objects, not able to represent events
 - Not description on syntax / semantic of a sentence, etc “bank”



4. The following statements are given.

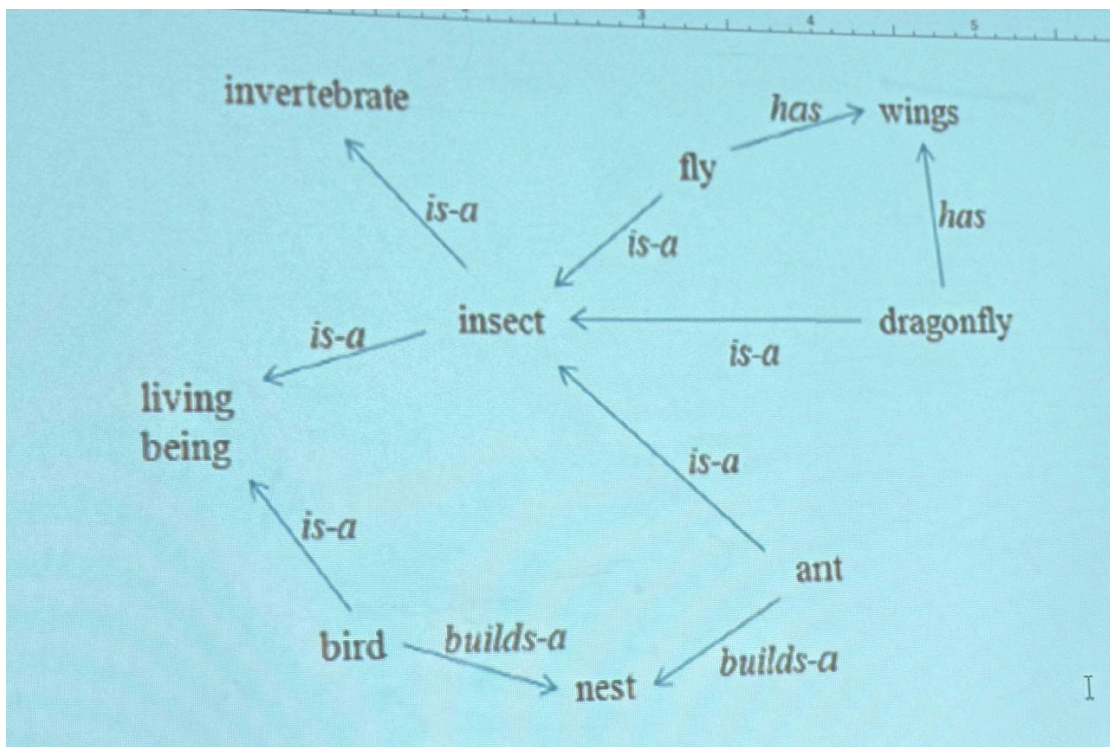
Bird is a living being that builds nest.
Insect is an invertebrate living being.
Ant is a kind of insects that builds nest.
Both dragonfly and fly are insects that have wings.

- a) Between semantic network and frames, justify which knowledge representation tool you would use to represent the information above. ()
- b) Discuss **ONE (1)** limitation of the knowledge representation tool that is selected in Question 4(a) above. ()
- c) Illustrate the representation of the given statements above using the knowledge representation tool that you selected in Question 4(a) above. ()

Answer in the below is not enough for the exam so need more explanation

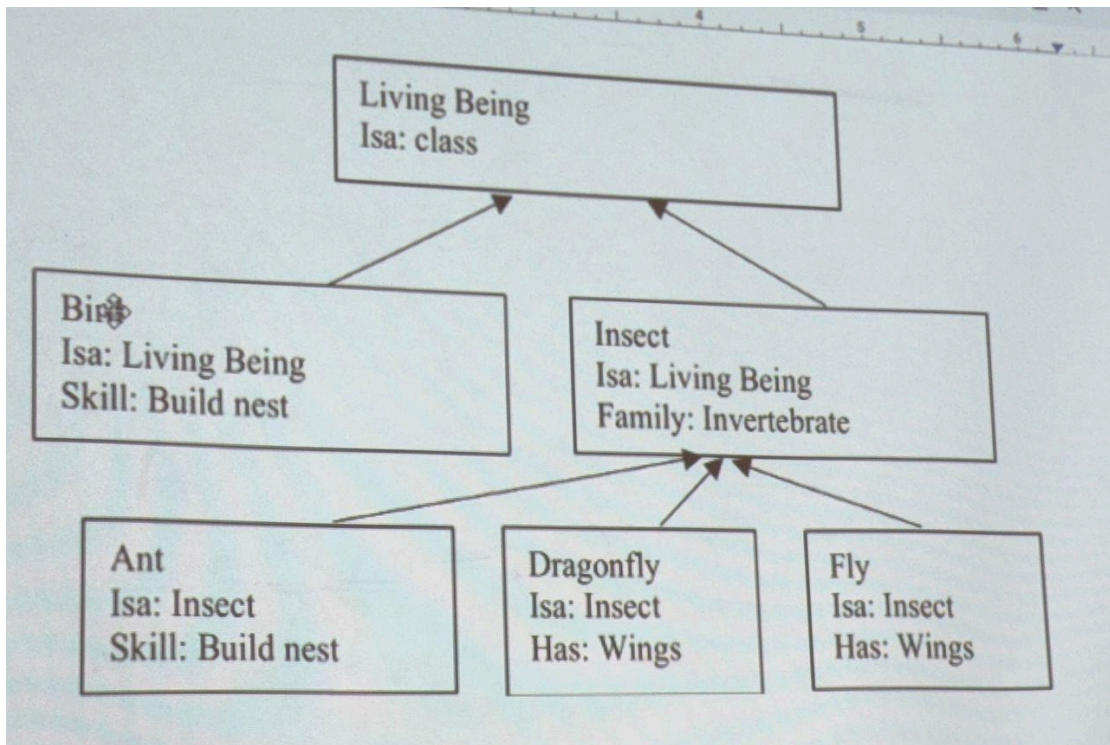
Semantic Network

- a) Semantic network as it is more suitable to describe the relationship between objects and overall concepts
- b) Limitation:
 - lack of standardization, no standards about node and arc values
 - difficult to view the relationship between a rule and another



Frame

- c) Frames as frames are suitable to show the details of objects and their relationships
- d) Limitation:
 - Only able to represent stereotypes objects, not able to represent events
 - Not description on syntax / semantic of a sentence, etc “bank”



5. Suggest and describe a real-life application where semantic network is applied. ()

WordNet

A real-life application is WordNet, a database that groups English words into sets of synonyms and connects them based on meaning. It shows how words relate to each other-like parts of something, opposites, or categories. WordNet is often used in search engines and language-related projects to help computers understand word meanings.

For example, the word is connected to "vehicle" (its hypernym), and "wheel" (a meronym).

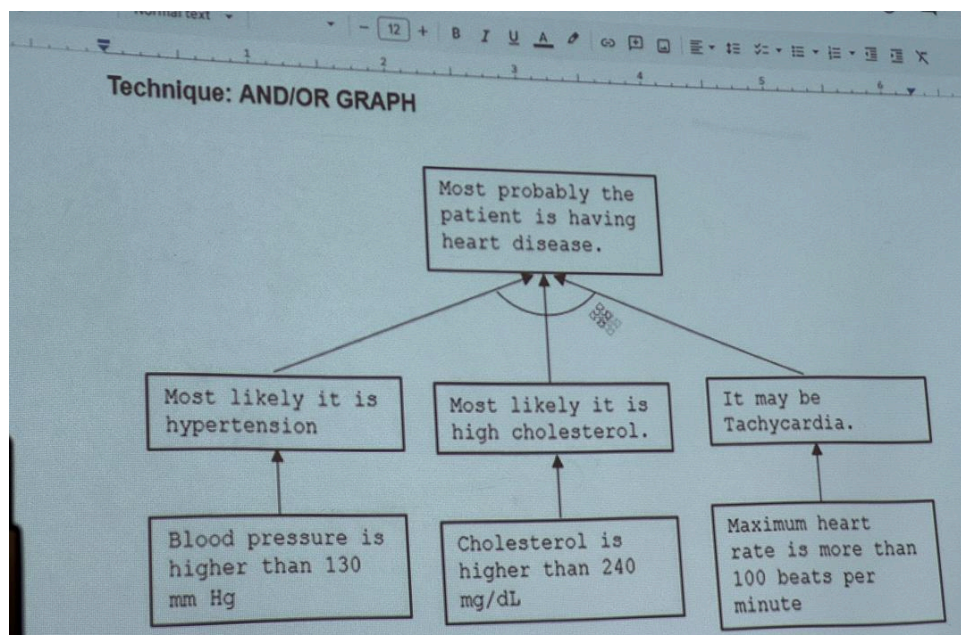
Google search engine

A real-life example of a semantic network is Google Search, especially its Knowledge Graph.

6. A rules based heart failure rule-based system is shown below.

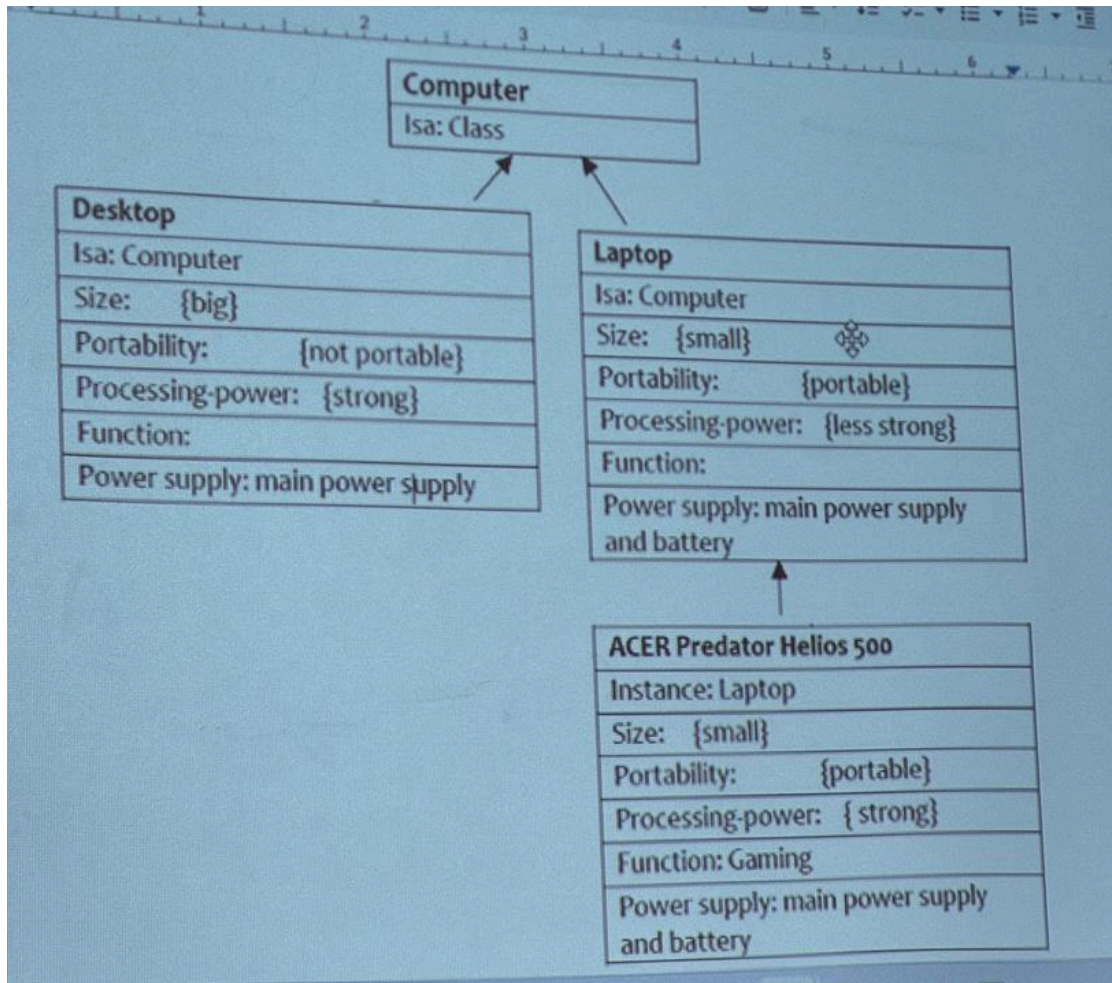
- R1: IF Blood pressure is higher than 130 mm Hg
THEN Most likely it is hypertension.
- R2: IF Cholesterol is higher than 240 mg/dL
THEN Most likely it is high cholesterol.
- R3: IF Maximum heart rate is more than 100 beats per minute
THEN It may be Tachycardia.
- R4: IF There is hypertension
AND High cholesterol
AND Tachycardia
THEN Most probably the patient is having heart disease.

Proposed **ONE (1)** technique to represent the knowledge above. Then, represent the knowledge by using the proposed technique. ()



7. The information below describes the differences between a desktop and a laptop. Illustrate the information below with frames. ()

Laptop and desktop are the commonly used computer among students. Laptop is generally small and easily portable. It can run either on battery or main power supply. Desktop, on the other hand, is large and not portable. It only runs on main power supply. Besides that, desktop normally has more powerful processor as compared to laptop. ACER Predator Helios 500, which is a gaming laptop, however, is equipped with a powerful Intel i9 processor.



Tutorial 6

1. Discuss the **THREE (3)** major obstacles involved in Natural Language Understanding (NLU). ()

Answer:

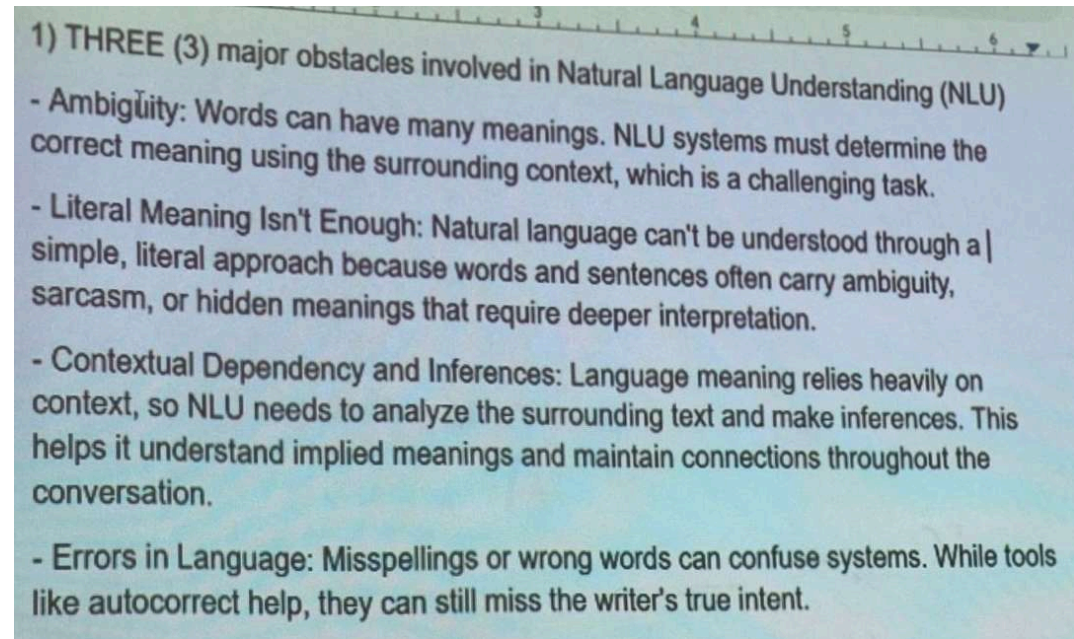
Transmission of words

Inferences about speaker's goals

Knowledge

Assumptions

The context of the interaction



2. What are the three typical stages involved in a Natural Language Processing (NLP) application development? Elaborate these three stages with the aid of appropriate diagrams. ()

Answer:

1. Parsing

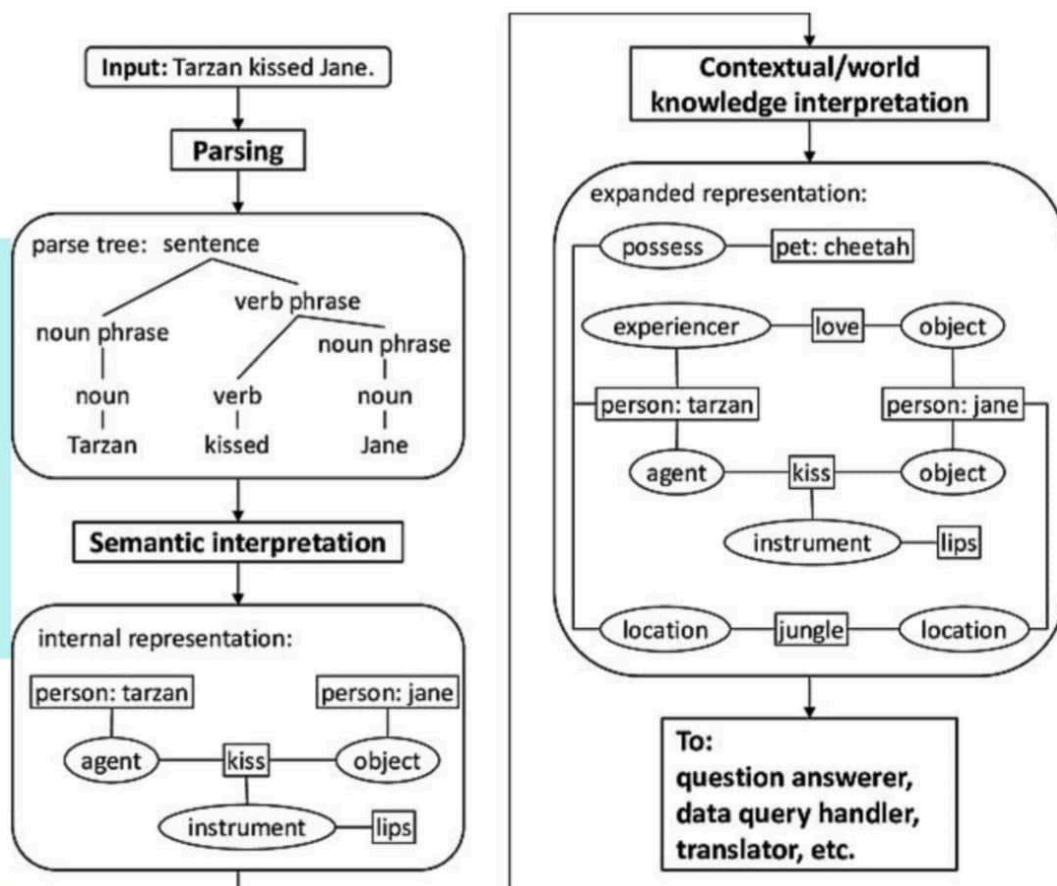
- Analyse the syntactic structure of sentences
- Identifying the major relations such as subject-verb, verb object, noun-modifier
- Often represented as parse tree
- Employs knowledge of language syntax, morphology, and some semantics

2. Semantic Interpretation

- Representation of the meaning
- Use semantic network, conceptual dependencies, frames, etc.
- Uses knowledge about the meaning of words and linguistics. Eg. The meaning of kiss is added "lips" as the instrument
- Perform consistency check and include constraints. EG. "kiss" can be performed by person, but not tree

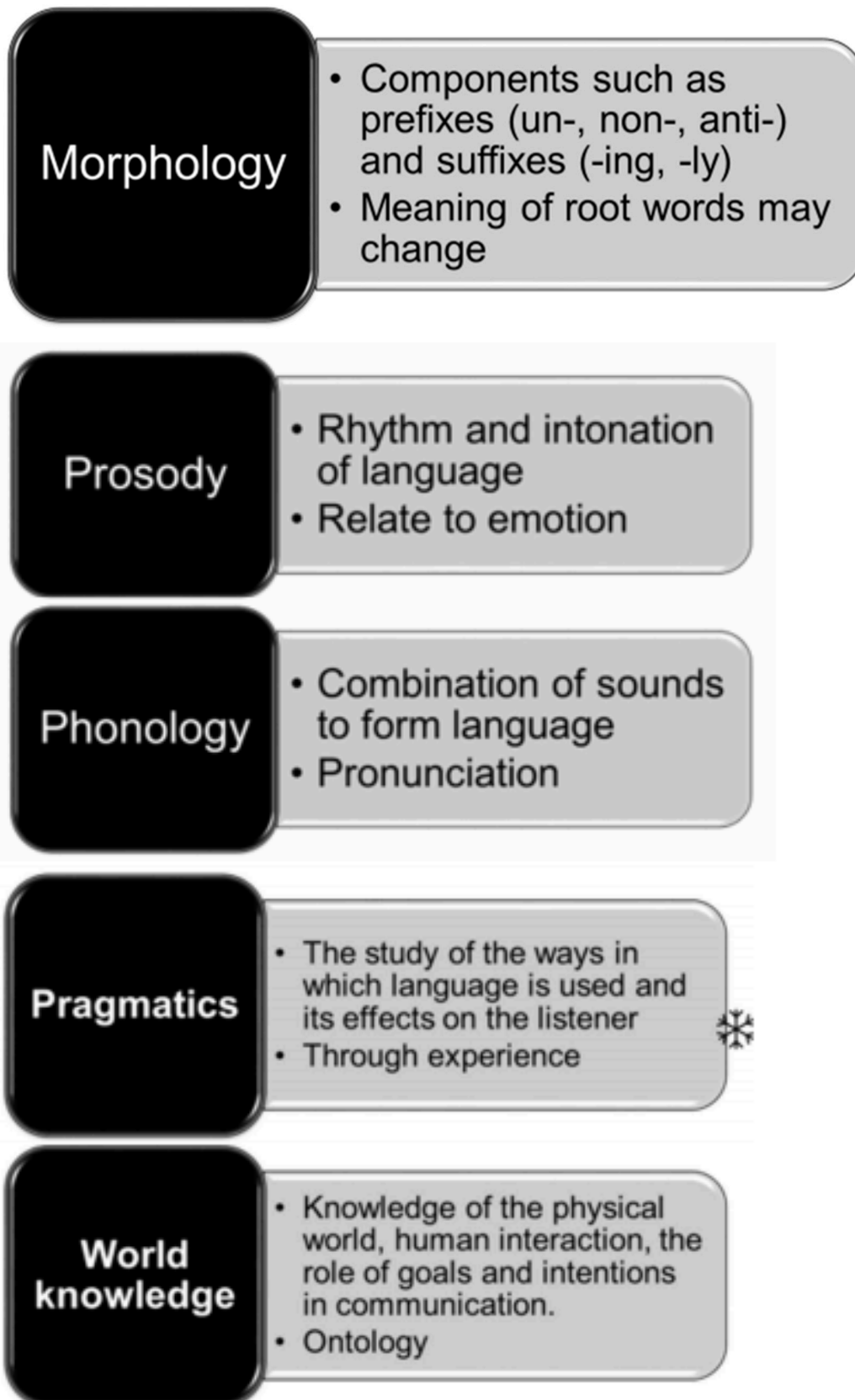
3. World knowledge representation

- Expanded representation of the sentence's meaning and added to the knowledge base.
- Necessary for complete understanding. Eg. Tarzan loves Jane, they live in jungle, Tarzan has pet Cheetah
- Used for further processing such as handling database queries, answering questions, translating the meaning, etc.



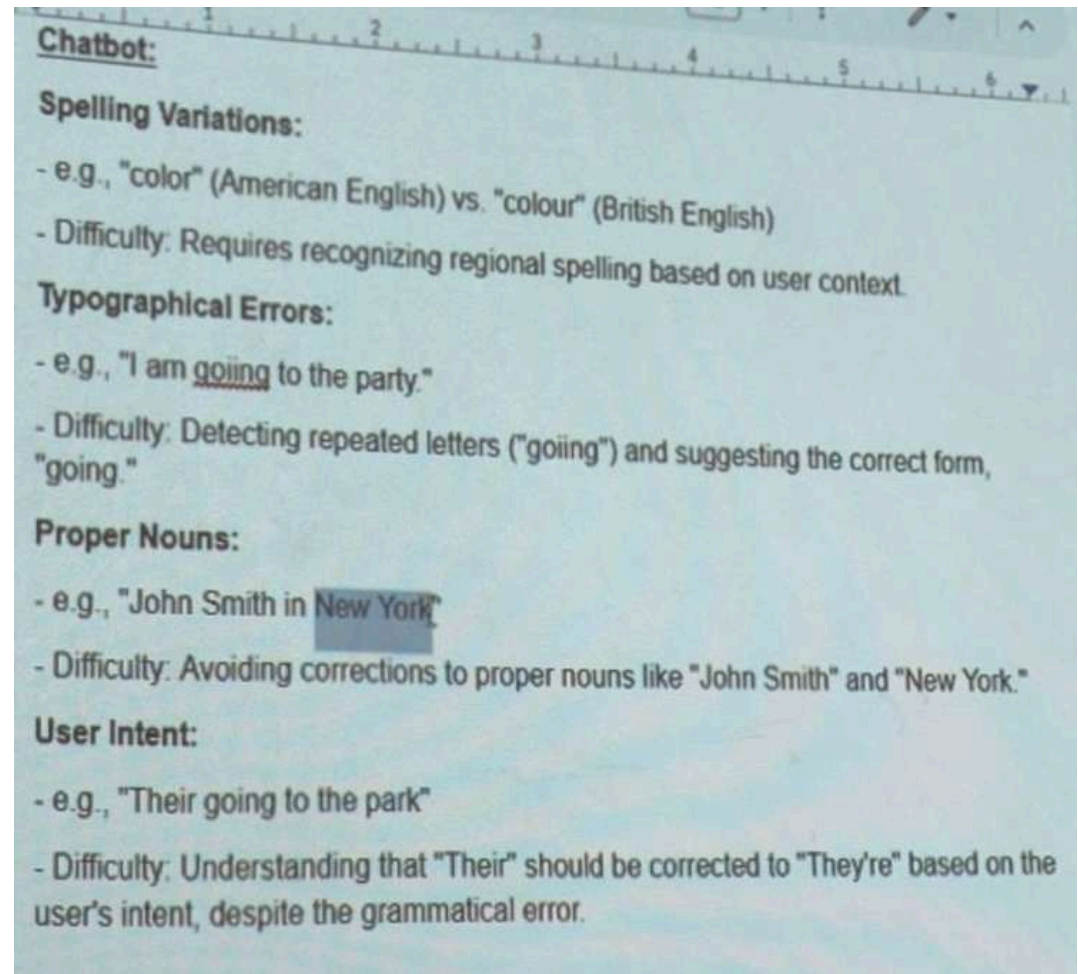
3. Besides the three main phases involved in NLP as discussed in Question 2, describe three other significant analyses that can be done to improve NLP applications. ()

Answer:



4. NLP application requires the use of knowledge about human languages. Suggest an example of an NLP application and describe the difficulties that a researcher would face during the development of the application that you have suggested. ()

Answer:



Chatbot:

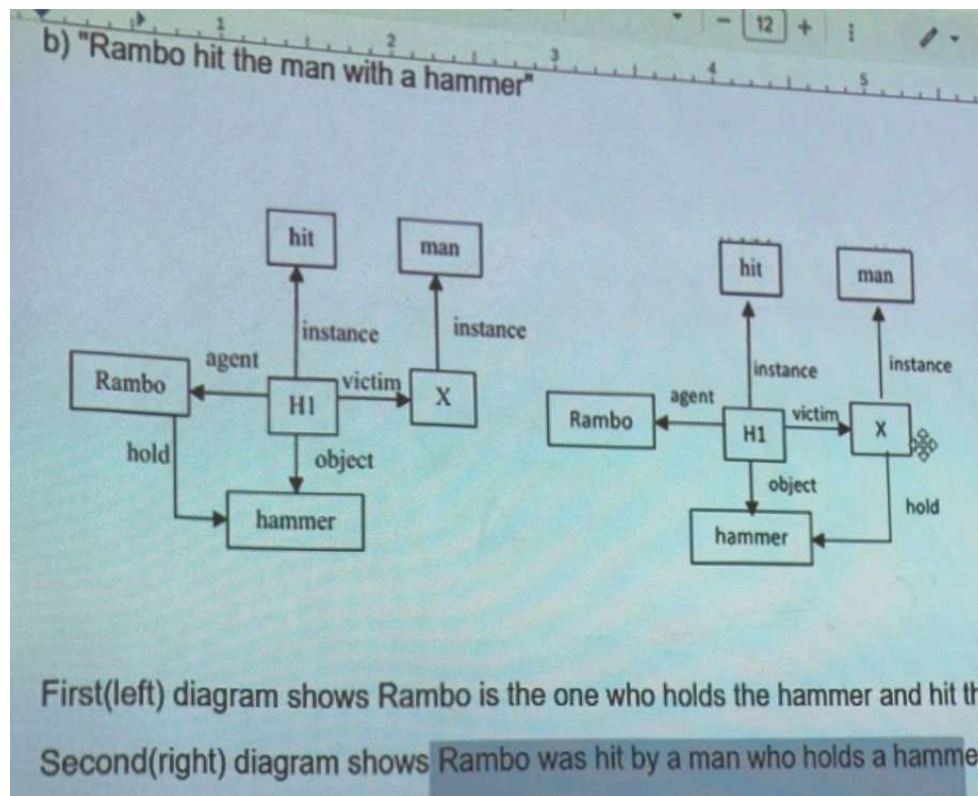
- Spelling Variations:**
 - e.g., "color" (American English) vs. "colour" (British English)
 - Difficulty: Requires recognizing regional spelling based on user context.
- Typographical Errors:**
 - e.g., "I am goiing to the party."
 - Difficulty: Detecting repeated letters ("goiing") and suggesting the correct form, "going."
- Proper Nouns:**
 - e.g., "John Smith in New York"
 - Difficulty: Avoiding corrections to proper nouns like "John Smith" and "New York."
- User Intent:**
 - e.g., "Their going to the park"
 - Difficulty: Understanding that "Their" should be corrected to "They're" based on the user's intent, despite the grammatical error.

5. In NLP, representation is important as it can solve issues like canonical form of sentences and syntactic problem of a sentence.
- Explain the meaning of canonical form of sentences. Provide examples to elaborate your answer.
 - By referring to the below statement, there is a syntactic problem. Identify the problem and then draw **TWO (2)** different representation (by selecting either the semantic network or the conceptual graph) to solve the syntactic problem. ()

"Rambo hit the man with a hammer"

Answer:

- Canonical form of sentences means two different sentences with the same meaning. For example, the sentences "I was given this hammer by Thor" and "Thor gave me this hammer" and



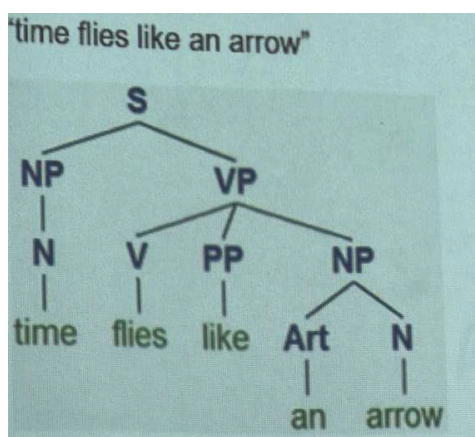
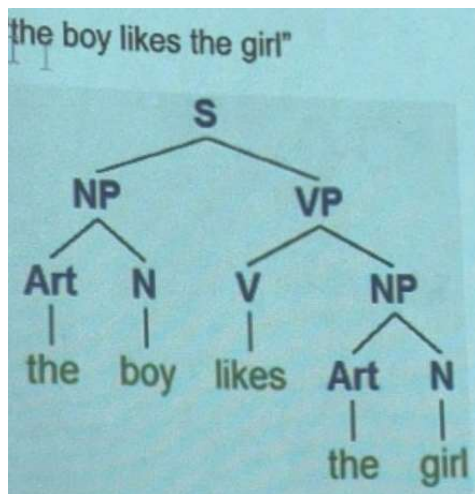
b)

6. Parse tree is a popular tool used in one of the phases of NLP called parsing. ()
- Explain the importance of parsing in NLP.
 - With the aid from the simple English grammar for simple transitive sentence as shown next page, draw the respective **parse trees** to verify the sentences "**the boy likes the girl**" and "**time flies like an arrow**"

Sentence	-> noun_phrase verb_phrase
Noun phrase	-> noun
Noun-phrase	-> article noun
Verb phrase	-> verb noun_phrase
Verb phrase	-> verb preposition noun_phrase
preposition	[like]
article	[a, an, the]
noun	[flies, time, arrow, boy, girl]
verb	[like, flies]

Answer:

- To analyse the syntactic structure of sentences
- To identify the major relations such as subject-verb verb-object noun modifier

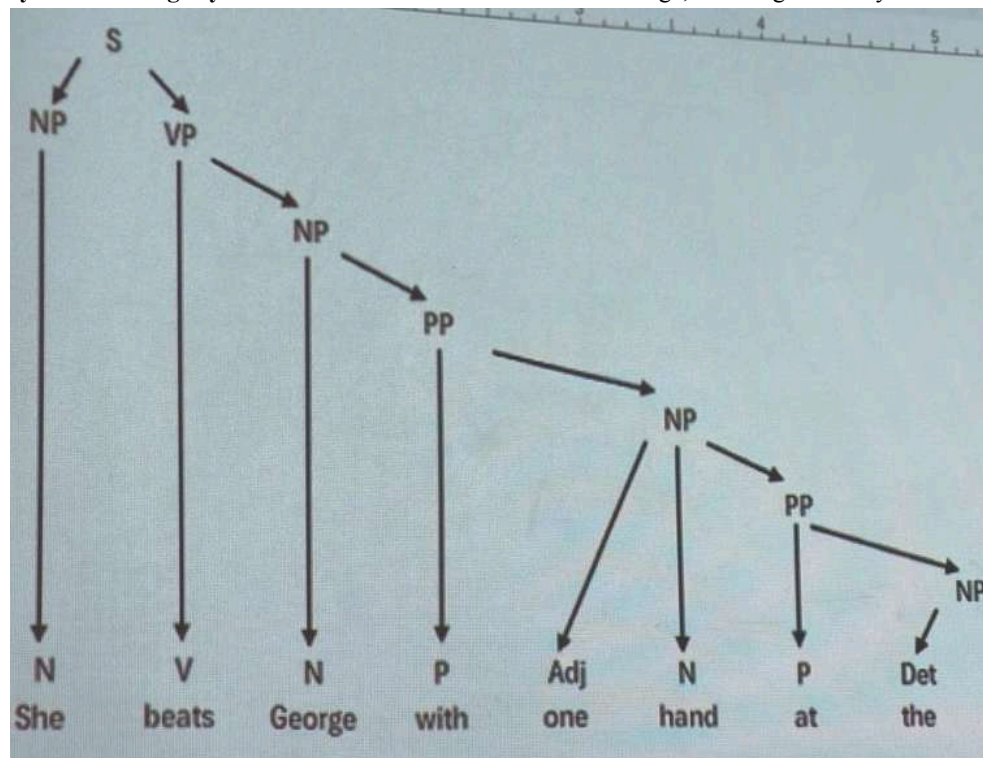


7. Considering the sentence S = "She beats George with one hand at the bank".()
- The sentence S consists of semantic ambiguity and syntactic ambiguity. Identify both of the ambiguities found from the sentence above.
 - Given the grammar below, construct **ONE (1)** parse tree for the sentence S. (**Remark:** grammar in the parentheses () means it is optional.)

S → NP VP (PP)
NP → (DET) (ADJ) N (PP)
VP → V NP (PP)
PP → P NP
ADJ → [one]
DET → [a, the]
N → [She, George, hand, bank]
V → [beats]
P → [with, at]

Answer:

- Semantic ambiguity:** bank means different things. Eg. A river bank or a financial institution
Syntactic ambiguity: It is either she uses one hand to beat George, or George has only 1 hand



b)

8. Consider a sentence, $S = \text{Ethan invited the person with microphone.}$ ()
- The sentence S consists of syntactic ambiguity. Identify the ambiguity found from the sentence above.
 - Given the grammar below, construct **ONE (1)** parse tree for the sentence S .

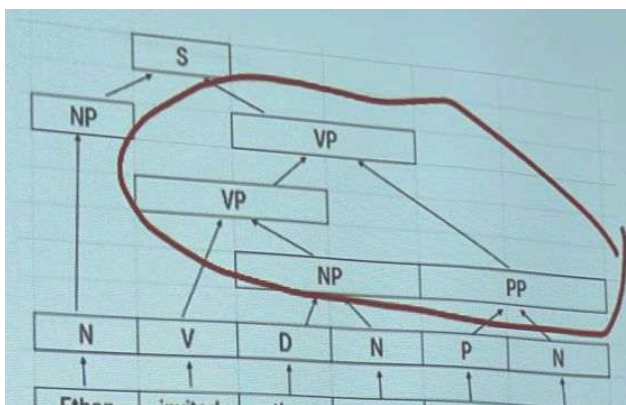
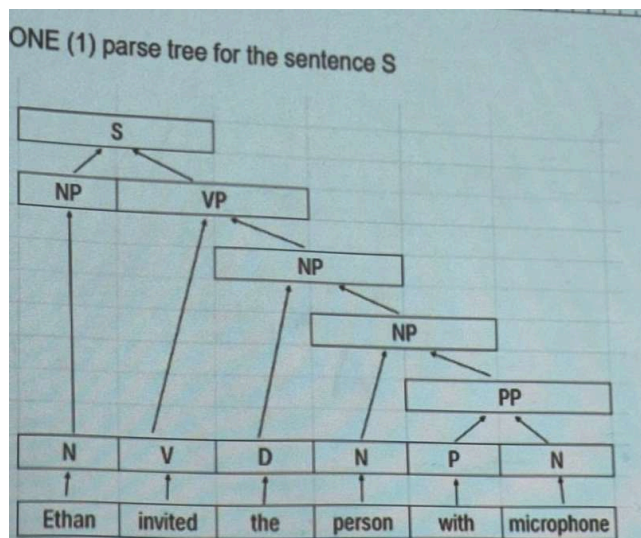
```
noun_phrase (NP)
verb_phrase (VP)
preposition_phrase (PP)
determiner (D)
noun (N)
verb (V)
S -> NP VP
NP -> N | N PP | D NP | D N
VP -> VP PP | V NP
PP -> P N
N -> [Ethan, person, microphone]
V -> [invited]
P -> [with]
D -> [the]
```

Answer:

- a) Syntactic ambiguity

Meaning 1: Ethan with microphone

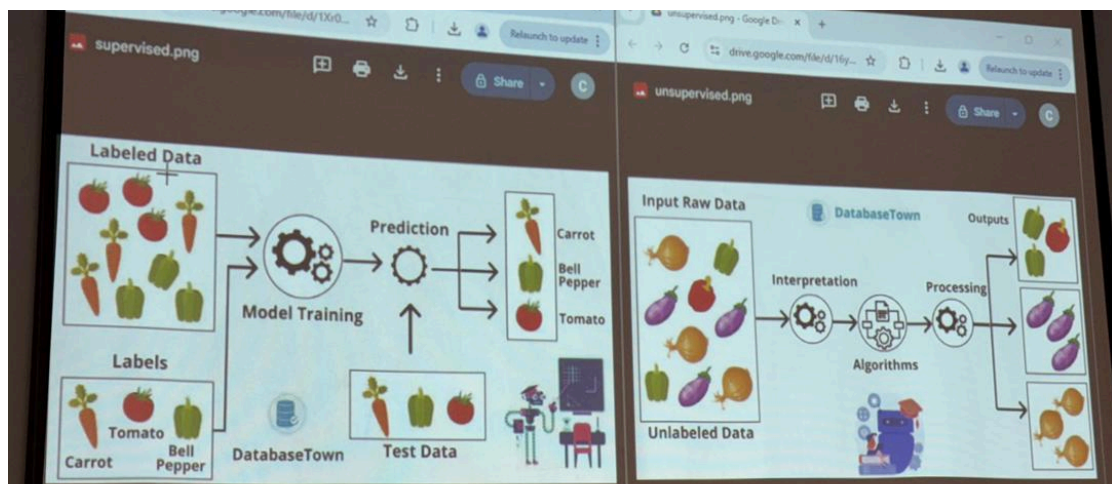
Meaning 2 : The person with microphone



Tutorial 7

1. Differentiate between artificial intelligence (A.I.) and machine learning (ML).

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	AI refers to the simulation of human intelligence in machine. It involves the ability of machines to perform tasks that normally require human intelligence.	ML is a subset of AI that involves the development and algorithms and models that allow computers to learn patterns from data and make decisions based on that learning.
Scope	AI is a broader concept that aims to create machines capable of mimicking human intelligence across a wide range of tasks.	ML focuses specifically on the development of algorithms that allow computers to learn from and make predictions or decisions based on data.
Goal	The goal of AI is to create machines that can replicate human-like intelligence and decision-making.	The goal of ML is to enable computers to learn from data, improve their performance over time, and make predictions or decisions based on that learning.
Type of Tasks	AI can perform tasks that include reasoning, problem-solving, decision-making, language understanding, and more.	ML is commonly used for tasks like classification, regression, clustering, and pattern recognition.



2. There are **TWO (2)** main types of ML which are Supervised and Unsupervised Learning. ()
- a) Discuss the differences between them.
- b) Provide **TWO (2)** algorithms for each type ML.

a)

Aspect	Supervised Learning	Unsupervised Learning
Definition	Supervised Learning involves training a model using labeled data, where the input data is paired with corresponding correct output or target values. The goal is to learn a mapping from inputs to outputs.	Unsupervised learning involves training a model on unlabelled data, where the algorithm tries to find patterns or structures within the data without specific target values.
Learning process	The model learns from examples provided in the form of input-output pairs. It learns to generalize and make predictions on new, unseen data.	The model learns to identify patterns, clusters, or relationships in the data without being explicitly given the correct answers.
Goal	The goal is to learn a mapping or relationship between inputs and outputs, enabling the model to predict the output for new, unseen inputs accurately.	The goal is to discover hidden patterns, groupings, or structure within the data, often for purposes like segmentation or anomaly detection.
Use Cases	Supervised learning is used for tasks where the correct output is known, and the model needs to generalize from the training data to make predictions.	Unsupervised learning is used for tasks where the goal is to explore and uncover insights from the data, often in cases where labelled data is unavailable.

b)

	Supervised Learning	Unsupervised Learning
Example Algorithms	Decision trees	K-means clustering
	Linear regression	Hierarchical Clustering

3. Assume that the Sea Fisheries Department has approached you to analyze abalone from physical measurements. A set of data that contains 7 attributes and 4177 rows of records is given. Part of the data are shown in **Table 1** below:

Table 1: Partial Abalone Data

Data Source: <http://mlr.cs.umass.edu/ml/datasets/Abalone>

Reference: Asuncion, A. & Newman, D.J. (2007). *UCI Machine Learning Repository*

[<http://www.ics.uci.edu/~mlearn/MLRepository.html>]. Irvine, CA: University of California, School of Information and Computer Science.

Between supervised learning or unsupervised learning, which type of ML is suitable to perform the following data analytics? Justify your answer. Then suggest **ONE (1)** suitable ML algorithm for each case.

- a) To determine 3 different grades of abalone based on diameter and whole weight. ()
- b) To predict the age of an abalone. (To determine the age of an abalone, we can check the number of rings, which is between 1 to 29) ()

Answer:

a) Unsupervised learning - to cluster based on the similar features
ML algorithm: K-means

b) Supervised learning - the output /label, i.e.. NUmber of rings is known
ML algorithm: Logistic Regression

4. Table 2 shows the partial data collection for stroke patients. The data is classified into two different classes which are Yes (1), and No (0). Identify the class that a new patient belongs to by using **K-Nearest Neighbour (K-NN) algorithm** given that the new patient's avg_glucose_level and bmi are 200.7 and 30.5, respectively (**Note:** K = 5). ()

Table 2: Attributes of the stroke dataset

patient_id	avg_glucose_level	bmi	stroke
9046	228.69	36.6	1
38047	100.98	28.2	1
56669	186.21	29.0	1
2374	213.37	36.0	0
15528	223.36	41.5	0
70374	122.41	40.3	0
1665	174.12	24.0	1
29908	103.26	25.4	0
11091	75.39	37.8	0
53882	70.09	27.4	1

Source: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset?resource=download>

Identify the class that a new patient belongs to by using **K-Nearest Neighbour (K-NN) algorithm** given that the new patient's avg_glucose_level and bmi are 200.7 and 30.5, respectively (**Note:** K = 5).

patient_id	avg_glucose_level	bmi	stroke	distance
9046	228.69	36.6	1	$\sqrt{((228.69 - 200.7)^2 + (36.6 - 30.5)^2)}$ $= \sqrt{(27.99^2 + 6.1^2)}$ $= \sqrt{(783.4401 + 37.21)}$ $= \sqrt{820.6501}$ ≈ 28.65
38047	100.98	28.2	1	$\sqrt{((100.98 - 200.7)^2 + (28.2 - 30.5)^2)}$ $= \sqrt{(-99.72^2 + (-2.3)^2)}$ $= \sqrt{(9944.0784 + 5.29)}$ $= \sqrt{9949.3684}$ ≈ 99.75
56669	186.21	29.0	1	$\sqrt{((186.21 - 200.7)^2 + (29.0 - 30.5)^2)}$ ≈ 14.57
2374	213.37	36.0	0	$\sqrt{((213.37 - 200.7)^2 + (36.0 - 30.5)^2)}$ ≈ 13.81
15528	223.36	41.5	0	$\sqrt{((223.36 - 200.7)^2 + (41.5 - 30.5)^2)}$ ≈ 25.19
70374	122.41	40.3	0	$\sqrt{((122.41 - 200.7)^2 + (40.3 - 30.5)^2)}$ ≈ 78.90
1665	174.12	24.0	1	$\sqrt{((174.12 - 200.7)^2 + (24.0 - 30.5)^2)}$ ≈ 27.36
29908	103.26	25.4	0	$\sqrt{((103.26 - 200.7)^2 + (25.4 - 30.5)^2)}$ ≈ 97.57
11091	75.39	37.8	0	$\sqrt{((75.39 - 200.7)^2 + (37.8 - 30.5)^2)}$ ≈ 125.52
53882	70.09	27.4	1	$\sqrt{((70.09 - 200.7)^2 + (27.4 - 30.5)^2)}$ ≈ 130.65

Among the 5 nearest neighbours:

Stroke = 0 -> 2 patients

Stroke = 1 -> 3 patients

P (Yes | new Patient) = $\frac{3}{5}$ = 0.6

P (No | new Patient) = $\frac{2}{5}$ = 0.4

Conclusion, the patient most probably belong to class "Yes"

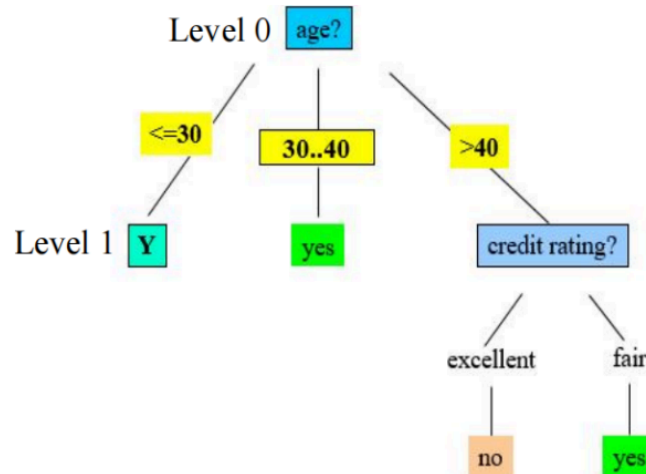


Figure 3: Partial decision tree of buys_computer

Table 2: The decision table of buys_computer

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

5. Given the table and the corresponding decision tree as above.
- Determine the appropriate attribute Y (income, student or credit_rating) at level 1 (Figure 3) by using Information Gain. Justify your answer. ()
 - Produce the complete decision tree based on the answer from Question 6(a). ()
 - Given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{excellent} \rangle$, predict the class it belongs to. ()
 - Refer to **Table 2**, given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{excellent} \rangle$, predict the class it belongs to, $P(C|X)$, using Naïve Bayes classification. ()

To calculate the information gain:

$$I(p, n) = -\frac{p}{p+n} \log_2 \frac{p}{p+n} - \frac{n}{p+n} \log_2 \frac{n}{p+n}$$

$$E(A) = \sum_{i=1}^v \frac{p_i + n_i}{p+n} I(p_i, n_i)$$

$$\text{Gain}(A) = I(p, n) - E(A)$$

a) From the table, for rows with **age <= 30**:

So total: 5 samples

Yes : 2 (2/5 = 0.4)

No : 3 (3/5 = 0.6)

Entropy of the parent node (age <= 30)

$$\begin{aligned}\text{Overall, } I(P, N) &= I(2, 3) = -(2/5) \cdot (\log_2(2/5)) - (3/5) \cdot (\log_2(3/5)) \\ &= -(0.4) \cdot (-1.32193) - (0.6) \cdot (-0.6737) \\ &= 0.971\end{aligned}$$

I(P, N) = I(2, 3) = so the P is 2 and N is 3

P is positive N is negative

$$I(p, n) = -\frac{p}{p+n} \log_2 \frac{p}{p+n} - \frac{n}{p+n} \log_2 \frac{n}{p+n}$$

$$E(A) = \sum_{i=1}^v \frac{p_i + n_i}{p+n} I(p_i, n_i)$$

$$\text{Gain}(A) = I(p, n) - E(A)$$

Income Attribute:

Income attribute:			
Income Level	P (Result with Yes)	N (Result with No)	I (P, N)
Low	1	0	$I(1, 0)$ $= -(1/1) \cdot (\log_2(1/1)) - (0/1) \cdot (\log_2(0/1))$ $= -(1) \cdot (0) - (0) \cdot (0)$ $= 0$
Medium	1	1	$I(1, 1)$ $= -(1/2) \cdot (\log_2(1/2)) - (1/2) \cdot (\log_2(1/2))$ $= -(0.5) \cdot (-1) - (0.5) \cdot (-1)$ $= 1$
High	0	2	$I(0, 2)$ $= -(0/2) \cdot (\log_2(0/2)) - (2/2) \cdot (\log_2(2/2))$ $= -(0) \cdot (0) - (1) \cdot (0)$ $= 0$

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$E(\text{Income_Level}) = (1/5) \cdot 0 + (2/5) \cdot 1 + (2/5) \cdot 0 = 0.4$$

$$\text{Gain}(\text{Income_level}) = 0.9710 - 0.4 = 0.5710$$

5 is sum up the total of result with yes and no

Student attributes:

Student attribute:			
Student	P (Result with Yes)	N (Result with No)	I (P, N)
No	0	3	$I(0, 3)$ $= 0$
Yes	2	0	$I(2, 0)$ $= 0$

$E(\text{Student}) = (3/5) \cdot 0 + (2/5) \cdot 0 = 0$

$\text{Gain}(\text{Student}) = 0.9710 - 0 = 0.9710$

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Credit Rating attributes:

Credit rating attribute:

Credit Rating	P (Result with Yes)	N (Result with No)	$I(P, N)$
Fair	1	2	$I(1, 2) \approx 0.9183$
Excellent	1	1	$I(1, 1) = 1$

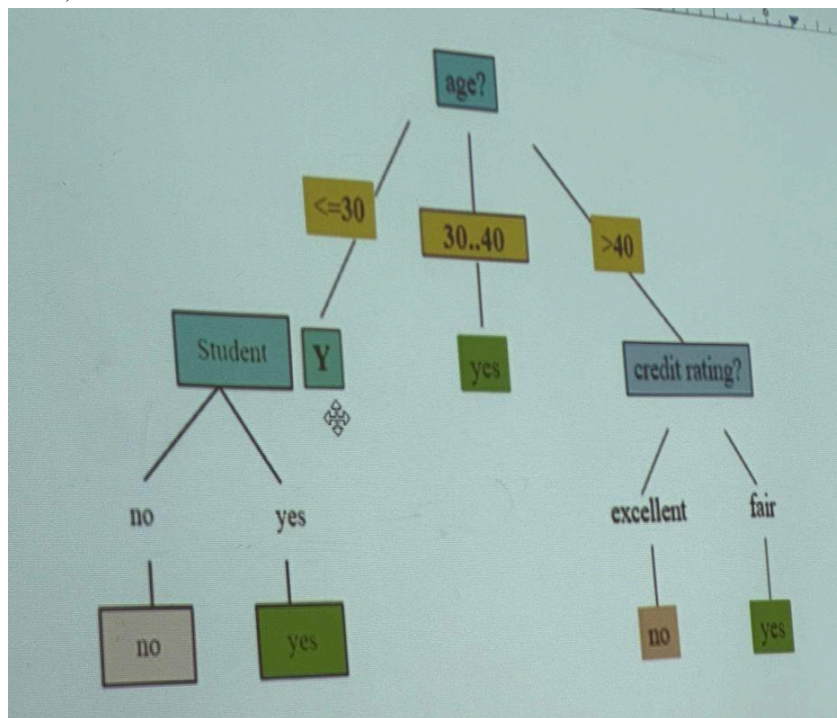
$E(\text{Credit_Rating}) = (3/5) * 0.9183 + (2/5) * 1 = 0.9510$
 $\text{Gain}(\text{Credit_Rating}) = 0.9710 - 0.95098 = 0.02$

The attribute "student" is selected as Y attribute as it has the highest

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

The attribute "student" is selected as Y attributes as it has the highest Information Gain

b)



c) Given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{excellent} \rangle$, predict the class it belongs to

Trace through the decision tree:

Age $\leq 30 \rightarrow$ go to student

Student = no $\rightarrow$ buys_computer = no

buys_computer = no

d) E

d) Given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{excellent} \rangle$, predict the class it belongs to, $P(C|X)$, using Naïve Bayes classification

$$P(C|X) = P(X|C) \cdot P(C) / P(X)$$

$P(p) = 9/14$
 $P(n) = 5/14$

$$P(X|p) \cdot P(p)$$

$$= P(\leq 30 | p) \cdot P(\text{low} | p) \cdot P(\text{no} | p) \cdot P(\text{excellent} | p) \cdot P(p)$$

$$= 2/9 \cdot 3/9 \cdot 3/9 \cdot 3/9 \cdot 9/14$$

$$= 0.00529$$

age	income	student	credit rating	buys computer
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
≥ 40	medium	no	fair	yes
≥ 40	low	yes	fair	yes
≥ 40	low	yes	excellent	yes
31...40	low	yes	excellent	no
≤ 30	medium	no	fair	yes
≤ 30	low	yes	fair	yes
≥ 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
≥ 40	medium	no	excellent	no

$$P(X|n) \cdot P(n)$$

$$= P(\leq 30 | n) \cdot P(\text{low} | n) \cdot P(\text{no} | n) \cdot P(\text{excellent} | n) \cdot P(n)$$

$$= 3/5 \cdot 1/5 \cdot 4/5 \cdot 3/5 \cdot 5/14$$

$$= 0.02057$$

The conclusion of buys_computer = no

6. Assess the following performance of the classifier as shown in Figure 4:
- Overall accuracy()
 - Precision for cat classification()
 - Recall for dog classification()

		Predicted		
		Cat	Dog	Rabbit
Actual class	Cat	5	3	0
	Dog	2	3	1
	Rabbit	0	2	11

Figure 4: Confusion Matrix

$$\text{Total} = 5+3+0+2+3+1+0+2+11$$

- a) Overall Accuracy

$$(\text{TP of Cat} + \text{TP of Dog} + \text{TP of Rabbit}) / \text{Total}$$

$$= (5+3+11) / (5+3+0+2+3+1+0+2+11)$$

$$= 19/27$$

$$= 0.7037$$

- b) Precision for cat classification

$$\text{TP of Cat} / \text{Predicated Yes}$$

$$= 5 / (5+2+0)$$

$$= 5/7$$

$$= 0.7143$$

- c) Recall for dog classification

$$\text{TP of Dog} / \text{Actual yes}$$

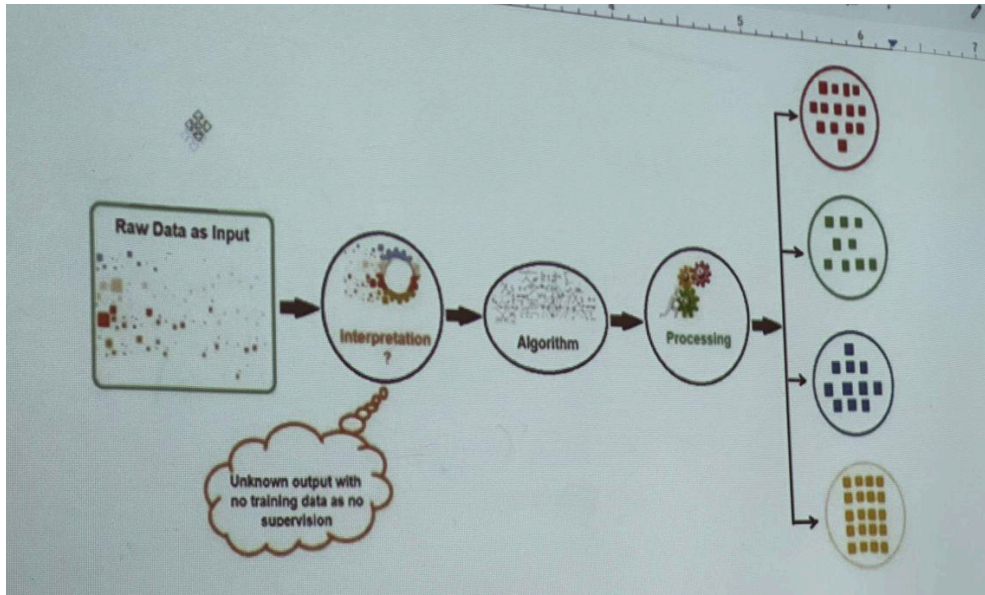
$$= 3 / (2+3+1)$$

$$= 3/6$$

$$= 0.5$$

Tutorial 8

1. Draw a diagram to depict an unsupervised learning model. ()



1. **Raw Data as Input:** The system starts with raw, unprocessed data, shown as scattered squares in different colors to represent the variety in data points.
2. **Interpretation:** This step involves understanding or interpreting the data.
3. **Algorithm:** The data is then processed through an algorithm. The data is processed using a set of rules or steps to transform, analyze, or group it.
4. **Processing:** This is the execution phase where the algorithm processes the data.
5. **Output:** Finally, the output is displayed as colored grids. Each grid has a uniform color and arrangement, indicating that the algorithm has organized the diverse raw data into distinct categories or results.

2. Considering a partial dataset as shown in Table 1: ()
- Explain how a regression model is different from clustering.
 - Explain how regression model and clustering can be applied on the dataset below.

Table 1: Sample of data

Age	Education	Gender	Income
31	Doctoral	Male	150k
45	Bachelor's	Female	78k
26	Master's	Male	76k
...	...	...	...

a) How regression model is different from clustering

Regression predicts a continuous-valued output. It is a statistical model used to predict numeric data rather than labels. It can also identify distribution trends based on available or historical data.

Clustering, on the other hand, involves dividing a dataset into groups called clusters.

The goal is to organize the data so that points within the same cluster are very similar, while points in different clusters are quite different. It is used to determine groupings in unlabeled data.

- b) **Regression:** Predicting a person's income from their age, education is example of regression task

Clustering: Example is to group the users based on their age, income and education

3. Clustering is the grouping of unlabeled data, it can be used for knowledge discovery. Discuss about this and provide **TWO (2)** real world applications of clustering. ()

Discuss about knowledge discovery and provide TWO (2) real world applications of clustering.

Clustering allows us to discover knowledge from unlabeled data by finding associations among features and grouping a set of objects so that those in the same group (called a cluster) are more similar to each other than to those in other groups (clusters).

Two real-life applications:

- T-Shirt Size Categorization - Clustering helps generalize sizes into a few standard categories that fit the majority of people.
- Image Segmentation - Dividing an image into regions representing different objects or areas, such as sky, grass, and buildings, for easier analysis and processing.

4. K-means is used to cluster the data in Table 2. Randomly we choose two cluster centers ($k = 2$) for two clusters; $C1 = (1.0, 1.0)$ and $C2 = (5.0, 7.0)$.

Table 2: Sample of data

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

- List the clusters that each data will be assigned to. ()
- Identify the new centers for the first iteration. ()
- Suggest TWO (2) stopping conditions for K-means. ()
- Discuss the limitation of K-means. ()

the distance of each data towards center 1 (1.0, 1.0) and center 2 (5.0, 7.0)

Individual	Variable 1	Variable 2	To C1	To C2	C1	C2
1	1.0	1.0	$\sqrt{(1.0-1.0)^2 + (1.0-1.0)^2}$ $= \sqrt{(0^2 + 0^2)}$ $= \sqrt{(0+0)}$ $= \sqrt{0}$ ≈ 0	$\sqrt{(1.0-5.0)^2 + (1.0-7.0)^2}$ $= \sqrt{(-4.0)^2 + (-6.0)^2}$ $= \sqrt{(16+36)}$ $= \sqrt{52}$ ≈ 7.21	✓	
2	1.5	2.0	$\sqrt{(1.5-1.0)^2 + (2.0-1.0)^2}$ $= \sqrt{(0.5^2 + 1.0^2)}$ $= \sqrt{(0.25 + 1.0)}$ $= \sqrt{1.25}$ ≈ 1.12	$\sqrt{(1.5-5.0)^2 + (2.0-7.0)^2}$ $= \sqrt{(-3.5^2 + (-5.0)^2)}$ $= \sqrt{(12.25 + 25.0)}$ $= \sqrt{37.25}$ ≈ 6.10	✓	
3	3.0	4.0	$\sqrt{(3.0-1.0)^2 + (4.0-1.0)^2}$ ≈ 3.61	$\sqrt{(3.0-5.0)^2 + (4.0-7.0)^2}$ ≈ 3.61	✓	
4	5.0	7.0	$\sqrt{(5.0-1.0)^2 + (7.0-1.0)^2}$ $= \sqrt{(4.0^2 + 6.0^2)}$ $= \sqrt{(16.0 + 36.0)}$ $= \sqrt{52.0}$ ≈ 7.21	$\sqrt{(5.0-5.0)^2 + (7.0-7.0)^2}$ $= \sqrt{(0^2 + 0^2)}$ $= \sqrt{(0+0)}$ $= \sqrt{0}$ ≈ 0		✓
5	3.5	5.0	$\sqrt{(3.5-1.0)^2 + (5.0-1.0)^2}$ $= \sqrt{(2.5^2 + 4.0^2)}$ $= \sqrt{(6.25 + 16.0)}$ $= \sqrt{22.25}$ ≈ 4.72	$\sqrt{(3.5-5.0)^2 + (5.0-7.0)^2}$ $= \sqrt{(-1.5^2 + (-2.0)^2)}$ $= \sqrt{(2.25 + 4.0)}$ $= \sqrt{6.25}$ ≈ 2.5		✓
6	4.5	5.0	$\sqrt{(4.5-1.0)^2 + (5.0-1.0)^2}$ ≈ 5.32	$\sqrt{(4.5-5.0)^2 + (5.0-7.0)^2}$ ≈ 2.06		✓
7	3.5	4.5	$\sqrt{(3.5-1.0)^2 + (4.5-1.0)^2}$ ≈ 4.30	$\sqrt{(3.5-5.0)^2 + (4.5-7.0)^2}$ ≈ 2.92		✓

We obtain two clusters containing: {1,2,3} and {4,5,6,7}

b) new centers for the first iteration

C1: (1.0,1.0), (1.5,2.0), (3.0,4.0)

$$C1_new = (1/3(1.0+1.5+3.0), 1/3(1.0+2.0+4.0))$$

$$= (1/3(5.5), 1/3(7.0))$$

$$= (1.83, 2.33)$$

C2: (5.0,7.0), (3.5,5.0), (4.5,5.0), (3.5,4.5)

$$C2_new = (1/4(5.0+3.5+4.5+3.5), 1/4(7.0+5.0+5.0+4.5))$$

$$= (1/4(16.5), 1/4(21.5))$$

$$= (4.13, 5.38)$$

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

c) Two stopping conditions for K-means

- no (or minimum) re-assignments of data points to different clusters, or
- no (or minimum) change of centroids, or
- minimum decrease in the sum of squared error

d) limitation of K-means

- Very sensitive to the initial points
- Must manually choose K
- K-means has problems when clusters are of differing Sizes, Densities, and Non-globular shapes
- K-means has problems when the data contains outliers.

5. Mean Shift is one of the popular clustering algorithms: ()
a) Demonstrate the algorithm for Mean Shift

Algorithm of Meanshift (refer to the animation in lecture note)

1. Choose a search window (width and location)
2. Compute the mean of the data in the search window
3. Center the search window at the new mean location
4. Repeat until convergence

b) Discuss the advantages and disadvantages of Mean Shift. [slide 45]

Advantages:

- Does not assume number of clusters
- Just a single parameter (window size)
- Finds a variable number of modes (Local Maxima)
- Robust to outliers

Disadvantages:

- Output depends on window size
- Computationally expensive
- Does not scale well with dimension of feature space

6. Explain why cluster evaluation is a hard problem. Identify **TWO (2)** methods based on internal information that we can use to evaluate a clustering algorithm. ()

The quality of a clustering is very hard to evaluate because:

- We do not know the correct clusters
- Cluster center are arbitrary (unclear/randomly)

Methods based on internal information that can be used to evaluate a clustering algorithm

a) Intra-cluster cohesion (compactness)

- Cohesion measures how near the data points in a cluster are to the cluster centroid
- Sum of squared error (SSE) is a commonly used measure

b) Inter-cluster separation (isolation)

- Separation means that different cluster centroids should be far away from one another

Tutorial 9

1. Answer the following questions:

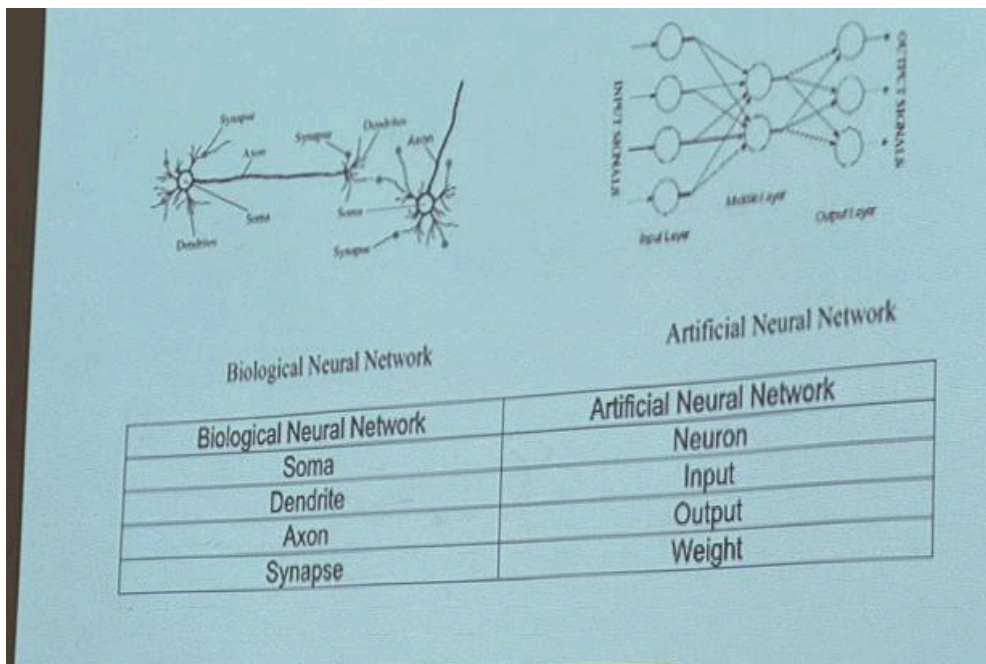
- a) Research in Artificial Neural Networks (ANNs) started with a view of mimicking the functioning of Biological Neural Networks (BNNs). Illustrate with diagrams, how the BNNs and ANNs are similar to each other. ()

A biological neuron consists of a soma (cell body), multiple dendrites, and a single axon. Connections between neurons form a biological neural network (BNN)

An artificial neuron consists of a neuron, a number of inputs with their correspondent adjusted weights, and an output. Connection between artificial neuron is called artificial neural network (ANN)

Similarities between BNN and ANN

- The processing element receives many signals
- Signals may be modified by a weight at the receiving synapse
- The processing element sums the weighted inputs
- The output from a particular neuron may go to many other neurons



- b) Figure 1 below shows the simplest neural network model that is known as perceptron, which can be used for classification tasks. Briefly explain the functions of each of the layers A, B and C as shown in the diagram. ()

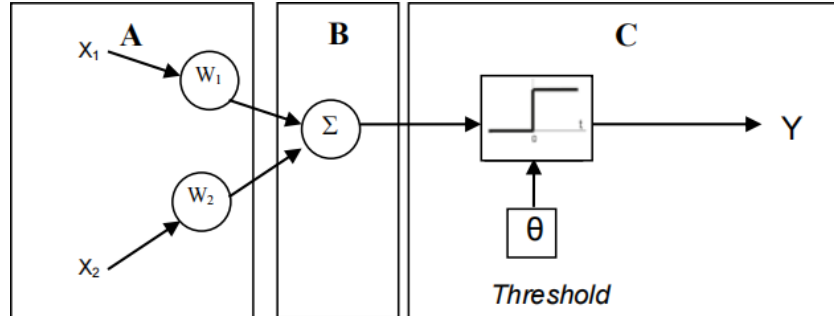
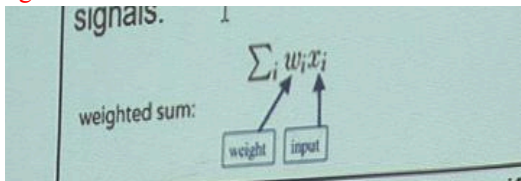


Figure 1: A sample neural network model

Layers	Functions	Elaborations
A	Input	Received data (x_1 and x_2) and the weights (w_1 and w_2) are used to express the strength or the importance of neuron input
B	Summation	The neuron computes the weights sum of the input signals 
C	Activation function	The step function is used for classification. If input $\geq$ threshold (θ), output =1; else output =0. The output can be the final classification or passed to another neuron

- c) Back propagation Multilayer Artificial Neural Network can be applied into a system that is able to recognize and interpret two different types of flowers. Discuss the prerequisite for the Neural Network system and design **ONE (1)** Neural Network structure for the purpose. State the advantage of back propagation network. ()

Back propagation Multilayer Artificial Neural Network can be applied into a system that is able to recognize and interpret two different types of flowers. Discuss the prerequisite for the Neural Network system and design ONE (1) Neural Network structure for the purpose. State the advantage of back propagation networks.

Training Data (Prerequisite):

- Collect flower features (e.g., petal length, petal width, sepal length, sepal width).
- Label each sample as Flower A or Flower B.

Neural Network Structure:

- Input layer: 4 nodes (flower features).
- Hidden layer: 5 nodes (learns patterns).
- Output layer: 2 nodes (Flower A / Flower B).

How Back Propagation Works:

- Network predicts flower type
- If prediction is wrong → error is calculated
- Weights are adjusted based on error, learning rate and epochs
- Repeats until error is minimized and classification improves

Advantages of Back Propagation network

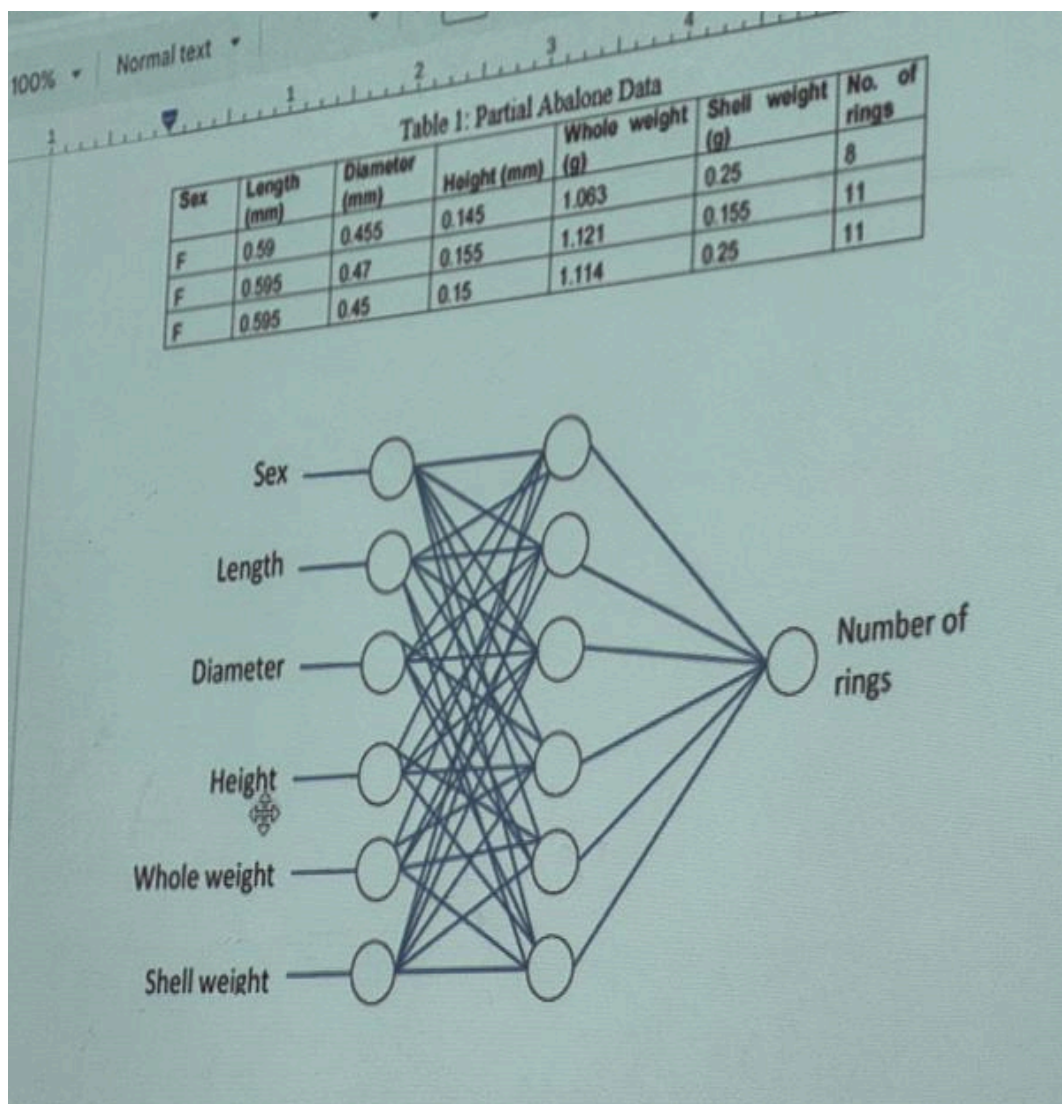
- Performs error correction: the network keeps fine tuning its weights to reduce mistakes, make it more accurate at recognizing the flower

2. Assume that the Sea Fisheries Department has approached you to analyze abalone from physical measurements. A set of data that contains 7 attributes and 4177 rows of records is given. Part of the data are shown in Table 1 below:

Table 1: Partial Abalone Data

Sex	Length (mm)	Diameter (mm)	Height (mm)	Whole weight (g)	Shell weight (g)	No. of rings
F	0.59	0.455	0.145	1.063	0.25	8
F	0.595	0.47	0.155	1.121	0.155	11
F	0.595	0.45	0.15	1.114	0.25	11

Demonstrate how a **multi-layer feedforward back-propagation neural network** can be constructed to predict the number of rings of abalone. Illustrate the architecture of the neural network with appropriate labels of each layer in the diagram. In your diagram, also include all the attributes and output as shown in Table 1. ()



3. A combinational logic circuit are made up from basic logic AND, OR, NAND, NOR or NOT gates that are “combined” or connected together to produce more complicated switching circuit. The output of the gate will be determined by combination of 1's and 0's which are present at the input terminals. The truth table of 3-input combinational logic circuit is given by Table 2 and Figure 2 where A, B and C are the inputs and Q is the output. ()

Table 2

A	B	C	Q
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

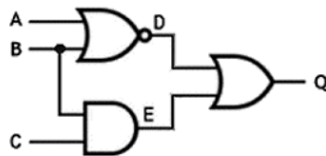


Figure 2

Train the perceptron to perform 3-input combinational logic circuit operation. The formula of output Q is given by

$$Q = w_1A + w_2B + w_3C$$

where w_1 , w_2 and w_3 are weight for A, B and C respectively. Assume that the threshold $\theta = 0.25$, learning rate $\alpha = 0.1$ and **step function** is used for activation function. Calculate the final weights of w_1 , w_2 and w_3 for **epoch 1 only**. (Assume the initial weights $w_1 = 0.3$, $w_2 = 0.2$ and $w_3 = 0.1$)

3) Train the perceptron to perform 3-input combinational logic circuit operation. The formula of output Q is given by

$$Q = w_1A + w_2B + w_3C$$

where w_1, w_2 and w_3 are weight for A, B and C respectively. Assume that the threshold $\theta = 0.25$, learning rate $\alpha = 0.1$ and step function is used for activation function. Calculate the final weights of w_1, w_2 and w_3 for epoch 1 only. (Assume the initial weights $w_1 = 0.3, w_2 = 0.2$ and $w_3 = 0.1$)

This type of activation function is called a **step function**.

Table 2

A	B	C	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$$\text{output} = \begin{cases} 0 & \text{if } \sum_i w_i x_i < \text{threshold} \\ 1 & \text{if } \sum_i w_i x_i \geq \text{threshold} \end{cases}$$

$$\text{output} = \begin{cases} 0 & \text{if } \sum_i w_i x_i + \text{bias} < 0 \\ 1 & \text{if } \sum_i w_i x_i + \text{bias} \geq 0 \end{cases}$$

$w_1 = 0.3, w_2 = 0.2, w_3 = 0.1, \theta = 0.25, \alpha = 0.1$

Actual output :

$$(A * W1) + (B * W2) + (C * W3)$$

Final Weight:

$$W1 + \alpha * A * e \text{ (e is the error)}$$

$$W2 + \alpha * B * e$$

$$W3 + \alpha * C * e$$

Then the first row final weight will be the second row initial weight.

Epoch	Inputs			Desired Output	Initial Weights			Actual Output	Error	Final Weights		
	A	B	C	Q _d	w ₁	w ₂	w ₃	Q	e	w ₁ + $\alpha * A * e$	w ₂ + $\alpha * B * e$	w ₃ + $\alpha * C * e$
1	0	0	0	1	0.3	0.2	0.1	$(0*0.3) + (0*0.2) + (0*0.1) = 0$ SINCE $0 < \theta (0.25)$ Actual Output = 0	Desired Output - Actual Output = $1 - 0 = 1$	$0.3 + 0.1 * 1 = 0.4$ $= 0.3$	$0.2 + 0.1 * 1 = 0.3$ $= 0.2$	$0.1 + 0.1 * 1 = 0.2$ $= 0.1$
	0	0	1	1	0.3	0.2	0.1	$(0*0.3) + (0*0.2) + (1*0.1) = 0.1$ SINCE $0.1 < \theta (0.25)$ Actual Output = 0	$1 - 0 = 1$	$0.3 + 0.1 * 0 = 0.3$	$0.2 + 0.1 * 0 = 0.2$	$0.1 + 0.1 * 1 = 0.2$
	0	1	0	0	0.3	0.2	0.2	0	0	0.3	0.2	0.2
	0	1	1	1	0.3	0.2	0.2	$(0*0.3) + (1*0.2) + (1*0.1) = 0.4$ SINCE $0.4 > \theta (0.25)$ Actual Output = 1	$1 - 1 = 0$	$0.3 + 0.1 * 0 = 0.3$	$0.2 + 0.1 * 1 = 0.3$	$0.2 + 0.1 * 1 = 0.3$
	1	0	0	0	0.3	0.2	0.2	1	-1	0.2	0.2	0.2
	1	0	1	0	0.2	0.2	0.2	1	-1	0.1	0.2	0.1
	1	1	0	0	0.1	0.2	0.1	1	-1	0.0	0.1	0.1
	1	1	1	1	0.0	0.1	0.1	0	1	0.1	0.2	0.2

4. In a supervised learning session, a 2-input neuron and the respective truth table are given in Figure 3 and Table 3, where x_1 and x_2 are the inputs and y is the output. ()

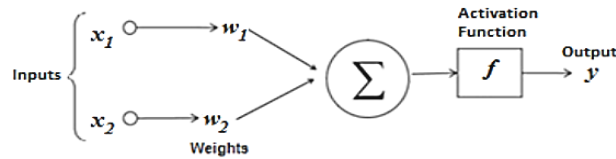


Figure 3: The 2-input neuron

Table 3

x_1	x_2	Desired Output, y_d
0	0	-1
0	1	1
1	0	1
1	1	1

The formula of output y is given by

$$y = \text{sign}(n) = \begin{cases} 1 & \text{if } n \geq \theta \\ -1 & \text{if } n < \theta \end{cases}$$

where $n = w_1x_1 + w_2x_2$, and w_1 and w_2 are the weights for x_1 and x_2 respectively. The **sign function** is used for activation function. The weights, $w_1 = 0.1$ and $w_2 = 0.2$, the threshold, $\theta = 0.2$ and the learning rate, $\alpha = 0.1$ are set initially for the training. Train the neuron and demonstrate the final weights of w_1 and w_2 for epoch 1 only.

Epoch		Inputs		Desired Output	Initial Weights		Actual Output	Error	Final Weights	
		x_1	x_2	y_d	w_1	w_2	y	e	w_1	w_2
1		0	0	-1	0.1	0.2	$(0 \times 0.1) + (0 \times 0.2) = 0$ SINCE $0 < \theta (0.2)$ Actual Output = -1	Desired Output - Actual Output $-1 - (-1) = 0$	$0.1 + 0.1 \times 0 \times 0 = 0.1$	$0.2 + 0.1 \times 0 \times 0 = 0.2$
		0	1	1	0.1	0.2	$(0 \times 0.1) + (1 \times 0.2) = 0.2$ SINCE $0.2 \geq \theta (0.2)$ Actual Output = 1	$1 - 1 = 0$	$0.1 + 0.1 \times 0 \times 0 = 0.1$	$0.2 + 0.1 \times 1 \times 0 = 0.2$
		1	0	1	0.1	0.2	-1	2	0.3	0.2
		1	1	1	0.3	0.2	1	0	0.3	0.2

Tutorial 10

1. Noise reduction is very important in image processing. ()

a) Discuss the above statement.

Digital images often contain noise, which happens when errors during image capture cause pixel values to differ from the real scene.

Noise reduction is important in the context of: (refer to slide page 34)

- To recover from the Image noise that might be caused by different intrinsic (i.e., sensor) and extrinsic (i.e., environment) conditions which are often not possible to avoid in practical situations.

- To ensure the smoothness and the best performance of the later processing steps.

- To eliminate unintended information during feature extraction.

b) Identify **TWO (2)** types of noises in an image. *[refer to slide 35]*

- Salt and pepper noise
- Gaussian noise
- Impulse noise

c) Name a filter to perform noise reduction.

Gaussian filter (gaussian noise), median filter (salt and pepper noise), and average filter (impulse noise)

- d) Demonstrate how image filtering works in reducing the noise by using a median filter. Perform median filtering onto the image in Figure below by using a 3×3 kernel. (Note: Replace the pixel value to 0 for border problems due to not enough neighbouring pixels to perform the filtering).

115	145	145	109	135	146
163	178	200	109	157	138
163	198	201	118	144	145
163	198	201	109	135	124
157	198	201	118	144	109
143	153	171	173	170	168
137	144	168	143	161	141

* Median filter = sorting from small to big *

* make sure the first pixel like 115 is in the middle of the 3×3 *

* have the border issues direct put 0 *

The image shows handwritten notes and diagrams illustrating the median filtering process. At the top, there are three boxes showing the sorting of pixel values for different 3×3 kernels and the resulting median values:

- Box 1: Sort: 115, 145, 145, 163, 178, 198, 200, 201. Median: 163
- Box 2: Sort: 109, 109, 118, 145, 178, 198, 200, 201. Median: 145
- Box 3: Sort: 109, 109, 118, 135, 138, 144, 145, 146, 157. Median: 138

Below these boxes, there are two 6×6 pixel grids. The left grid is the original image, and the right grid is the result of applying the median filter. The original image is:

115	145	145	109	135	146
163	178	200	109	157	138
163	198	201	118	144	145
163	198	201	109	135	124
157	198	201	118	144	109
143	153	171	173	170	168
137	144	168	143	161	141

The filtered image (right grid) shows the result of applying the median filter, with the median values (163, 145, 138) placed in the center of the 3×3 kernels. The border pixels are set to 0:

0	0	0	0	0	0
0	163	145	144	138	0
0	198	198	144	135	0
0	198	198	144	124	0
0	171	173	170	135	0
0	157	168	168	144	0
0	0	0	0	0	0

2. Morphological operation is important in image processing. ()
- a) Explain **FOUR (4)** morphological operations.
1. Erosion : makes white areas smaller and black area bigger
 2. Dilation : opposite of erosion, make white area grow bigger
 3. Opening :
 - 4.
- b) Suggest a morphological operation that could be used to recover the image as shown in figure below (the objective is to close the gap between the pixels of the number). Explain your answer.



Morphological operation: Dilation

Explanation: The dilation operation usually uses a structuring element for probing and expanding the shapes contained in the input image.

3. Explain **ONE (1)** importance of morphological image analysis in image processing. ()
- Useful to remove the noise
 - Improve image texture
4. In morphological image analysis, morphological operators often probe an image with a **structuring** element. A 6x6 image is given in **Figure 1** below. *(Note: Replace the pixel value to 0 for border problem due to not enough neighbouring pixels to perform the filtering)*

	0	1	2	3	4	5
0	0	0	0	0	0	1
1	0	1	1	1	1	1
2	0	1	1	1	1	0
3	0	1	1	1	1	0
4	0	1	1	1	0	0
5	1	1	1	0	0	0

Figure 1: 6x6 binary image

- a) Based on the 3x3 structuring element with the origin in the middle as shown in Figure 2, perform an **erosion** effect on the 6x6 binary image as shown in Figure 1. Provide the resulting binary image. ()

Erosion - all neighbour must be 1

1	1	1
1	1	1
1	1	1

Figure 2: 3x3 square structuring element

- b) Based on the 2x2 structuring element with the origin in the middle as shown in Figure 3, perform an **dilation** effect on the 6x6 binary image as shown in Figure 1. Provide the resulting binary image. ()

Dilation - at least 1 neighbour is 1

0	1
1	0

Figure 3: 2x2 square structuring element

Question a

$R' = R \ominus A$

Erosion = all neighbors must be 1.

0	0	0	0	0	1
0	<u>1</u>	<u>1</u>	<u>1</u>	1	1
0	<u>1</u>	<u>1</u>	<u>1</u>	1	0
0	<u>1</u>	<u>1</u>	<u>1</u>	1	0
0	1	1	1	0	0
1	1	1	0	0	0

0	0	0	0	0	0
0	0	0	0	0	0
0	0	<u>1</u>	<u>1</u>	0	0
0	0	<u>1</u>	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

Question b

$R' = R \oplus A$

Dilation = at least one neighbor is 1

<u>0</u>	<u>0</u>	0	0	0	1
<u>0</u>	<u>1</u>	1	1	1	1
0	1	1	1	1	0
0	1	1	1	1	0
0	1	1	1	0	0
1	1	1	0	0	0

0	0	0	0	0	0
0	<u>0</u>	1	1	1	1
0	<u>1</u>	1	1	1	1
0	1	1	1	1	1
0	1	1	1	1	0
0	1	1	1	0	0

Dilation = at least one neighbor is 1.

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0
0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0
0	0	1	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	0	0
0	0	0	0	0	1	1	1	0	0
0	0	0	0	0	1	1	1	0	0
0	0	0	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0

Thus, dilation is more suitable to restore the missing part of the Figure 4

6. Pre-processing is one of the important steps in digital image analysis. ()
- Explain the importance of image pre-processing.
 - Identify the suitable pre-processing operations based on the images in **Figure 7**. Provide brief description for each of the suggested operations.

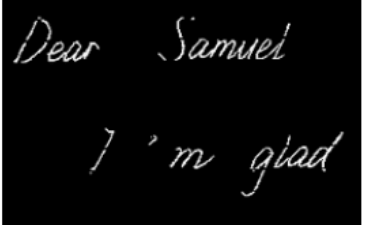
Image A (objects are less distinctive from its background)	Image B (salt and pepper in the image)	Image C (broken alphabet)
		

Figure 7: Sample images

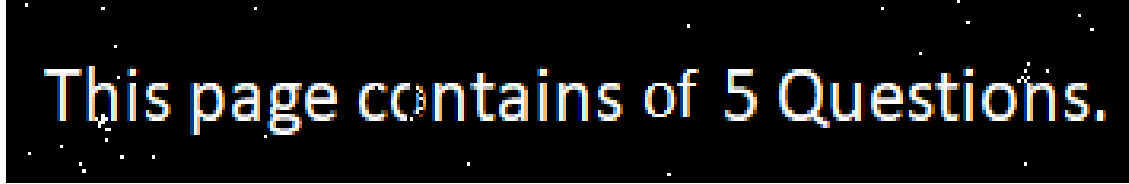
Question a

To improve the image quality (to increase chance for success of other processes)

Question b

- Contrast enhancement - to enhance the contrast of the image so that the intensity values of the image are sharpened.
- Noise removal - to remove noise from the image.
- Morphological operation - may perform dilation to fill up the broken strokes.

7. Figure below shows a part of the scanned image which will be recognized by optical character recognition (OCR) algorithm. Nevertheless, the detection performance is relatively poor compared to the actual sentence “This page contains of 5 Questions.” Identify TWO (2) potential output error produced by OCR algorithm. In an addition, provide ONE (1) preprocessing technique to increase the accuracy of OCR performance for each potential output error. ()



Potential error

1. “This” is detected as Tbis
2. “contains” is detected as ccntains
3. “Questions” is detected as Questiohs

Solution

1. Dilation or closing to enlarge the object to recover letter o in contains
2. Erosion / opening / noise remover to shrink the object to remove the noise for letter

Tutorial 11

Part I: Expert Systems

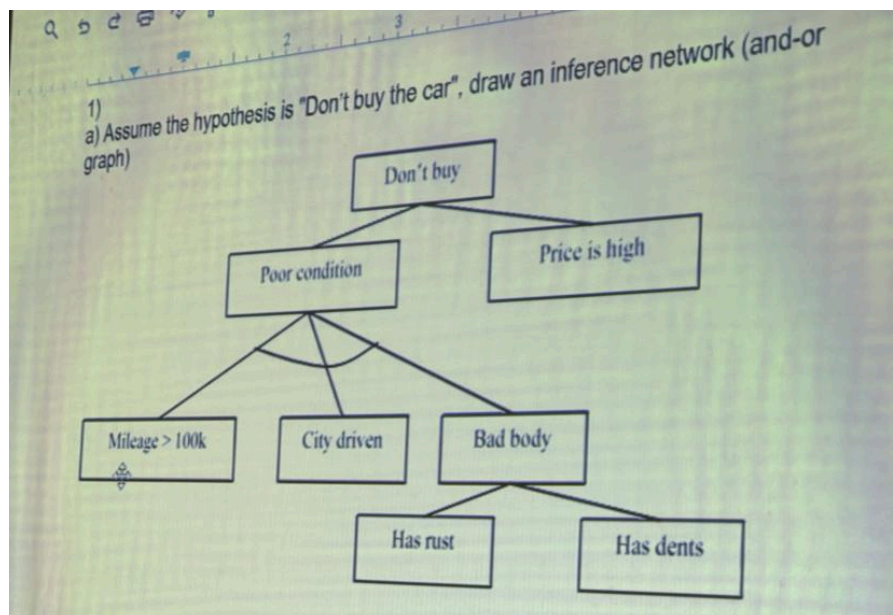
1. Given a customer has a problem deciding whether or not to purchase a car. He consulted an expert for advice and the consultant suggested some rules as follows.

- Rule 1: IF The condition of car is poor
 OR The price of the car is high
 THEN Don't buy the car
- Rule 2: IF Mileage on the car exceeds 100,000
 AND The car is city driven
 AND The body of the car is bad
 THEN The condition of the car is poor
- Rule 3: IF The car has dents
 THEN The body of the car is bad
- Rule 4: IF The car has rust
 THEN The body of the car is bad

- a) Assume the hypothesis is "Don't buy the car", draw an inference network (and-or graph) that supports the problem above. ()
- b) Given the observation by the expert as follows, demonstrate the inference process by using forward-chaining. ()

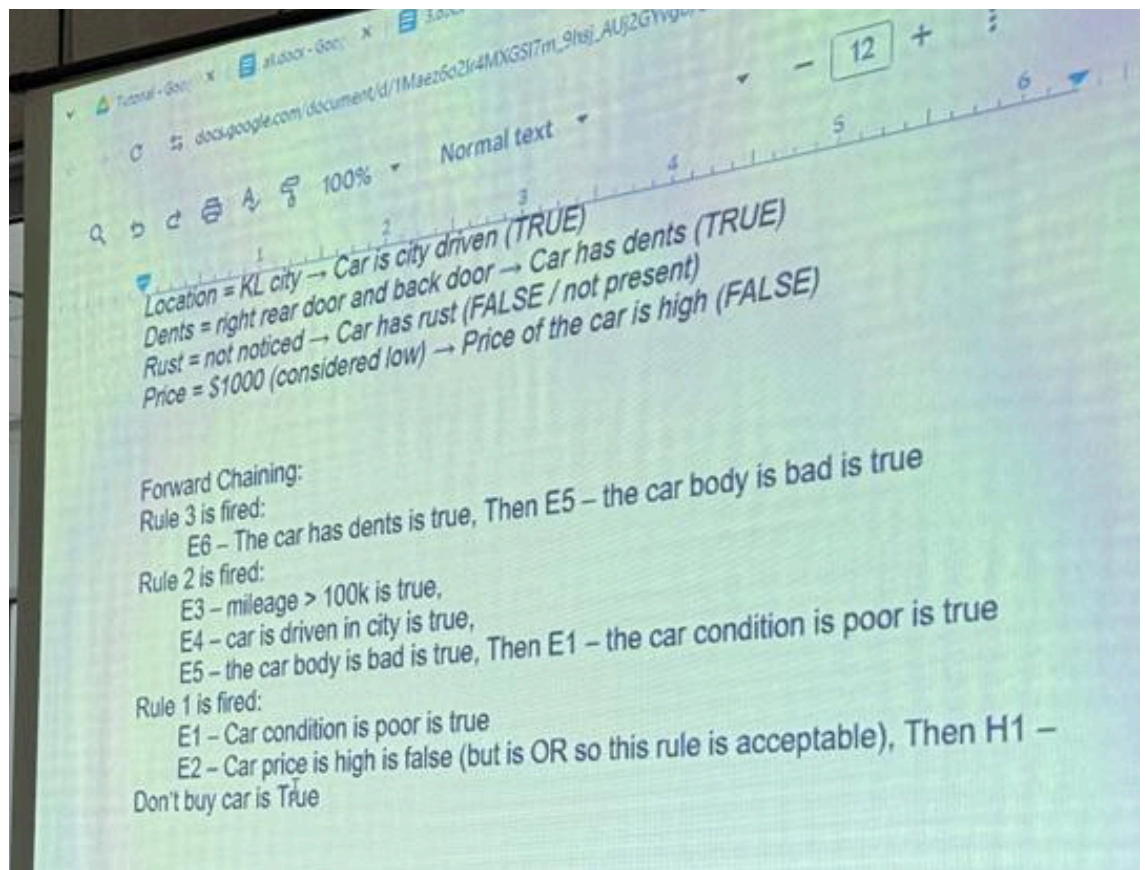
Mileage: 160,000 miles
Location of car: KL city
Dents on car: right rear door and back door
Rust on car: not noticed
Price of car: \$1000 (remark: considered low)

- c) If you are going to develop the above expert system, which inference mechanism are you going to implement? Forward or backward chaining? Justify your answer. ()



a)

b) Forward-chaining.



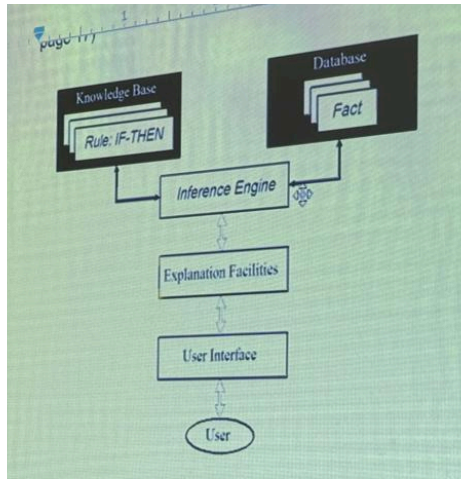
c) If you are going to develop the above expert system, which inference mechanism are you going to implement? Forward or backward chaining?

Forward chaining → Start from facts (mileage, dents, price) → apply rules → reach conclusion (don't buy).

Backward chaining → Start from hypothesis (don't buy) → work backwards → check/ask for needed facts.

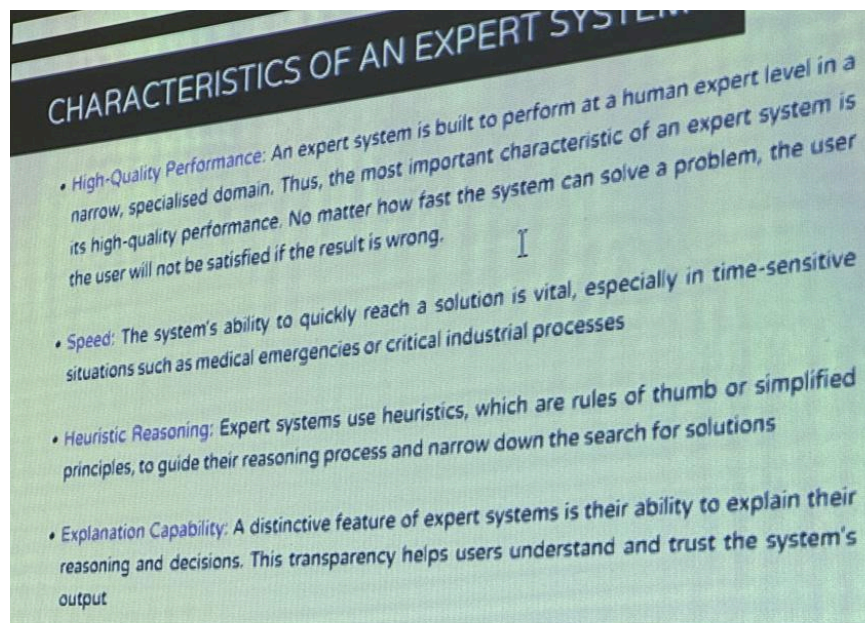
2. A basic structure of a rule-based expert system involves the components such as knowledge base, database, inference engine, explanation facilities, user interface, and user. ()
- Draw a diagram to illustrate the relations between the components in an expert system as given above, and describe the functions of each component.
 - Define the fundamental characteristics of an expert system.
 - What is the reason that an expert-system separates its knowledge base from the inference mechanism in the system?

a) Slide 17



- Knowledge Base: Contains domain knowledge or set of rules, will fire rules to execute necessary action
- Database: The set of facts will match with the rules in the knowledge base
- Inference Engine: Carry out the reasoning. May use forward-chaining, backward chaining & conflict resolution procedure.
- Explanation Facilities: Allow the user to ask the system how the conclusion is reached and why a specific fact is needed
- User Interface: Means of communication between user and the ES

b) Slide 18



- c) What is the reason that an expert-system separates its knowledge base from the inference mechanism in the system?
- Easy to maintain: new rules can be added without changing the inference engine, and vice versa.
 - Easy to understand: rules are readable without knowing how the inference mechanism works.

3. Given the rules below:

R1:	If X breathes using gills Then X lives in sea
R2:	If X has backbone and has fin and breathes using gills Then X is fish
R3:	If X lives in sea and has backbone and has fin Then X is seahorse
R4:	If X is fish and has fin and has two eyes at two sides Then X is shark
R5:	If X is fish and has two eyes at one side Then X is flounder

A subject with ID "X1001" was observed and some facts were gathered as listed below:

- ☐ X1001 has backbone
 - ☐ X1001 has fins
 - ☐ X1001 breathes through gills
 - ☐ X1001's two eyes on two sides of its head
- a) From the observation above, conflicts may occur when you infer the rules above. Analyze what is the potential conflict that might occur. ()
- b) Suggest and discuss **THREE (3)** approaches of conflict resolution. ()
- c) Derive a conclusion for the problem above for each of the conflict resolution approaches that you have suggested in Question 3(b) by using forward chaining. ()

a) Initially R1 And R2 can be fired therefore conflict is generated

b) Eg : [slide 23]

1. Highest priority
2. Most specific rule
3. Dat most recently entered
4. meta-rules
5. First come first serve

- **Most Specific Rule** - Fire the rule with the most detailed conditions.
Example: R2 is more specific than R1, so fire R2.
- **Highest Priority (Bottom most)** - Assign priorities to rules and fire the one with higher priority. Example: choose R2 over R1.
- **First Come First Serve** - Fire the earliest matched rule. Example: R1 fires first.

c)

- Most Specific rules: R2 is fired $\rightarrow$ R4 is fired (Conclusion: X1001 is shark)
- Highest Priority (Bottom most): R2 is fired $\rightarrow$ R4 is fired (Conclusion: X1001 is shark)
- First Come First Serve: R1 is fired $\rightarrow$ R2 is fired $\rightarrow$ R3 is fired $\rightarrow$ R4 is fired (Conclusion: X1001 is shark)

Part II: Recommender Systems

1. There are two common types of recommendation engine algorithms, i.e. collaborative filtering models and content-based models. Briefly explain the differences between them. Provide examples of applications to aid your answers. ()

[slide 52 & 54]

- Recommendation engines are mainly of two types: Collaborative Filtering and Content-Based.
- Collaborative Filtering: Uses user-item feedback (ratings, likes, or behavior such as watch time) to make predictions. It relies on patterns from many users.
- Content-Based Filtering: Uses item features (e.g., product price, color, movie genre, duration) to recommend similar items. Useful for cold-start problems when detailed descriptions are available.

2. Discuss the limitations of collaborative filtering models. (slide 55)

Cold Start - for new user or item, there isn't enough data to make accurate recommendations

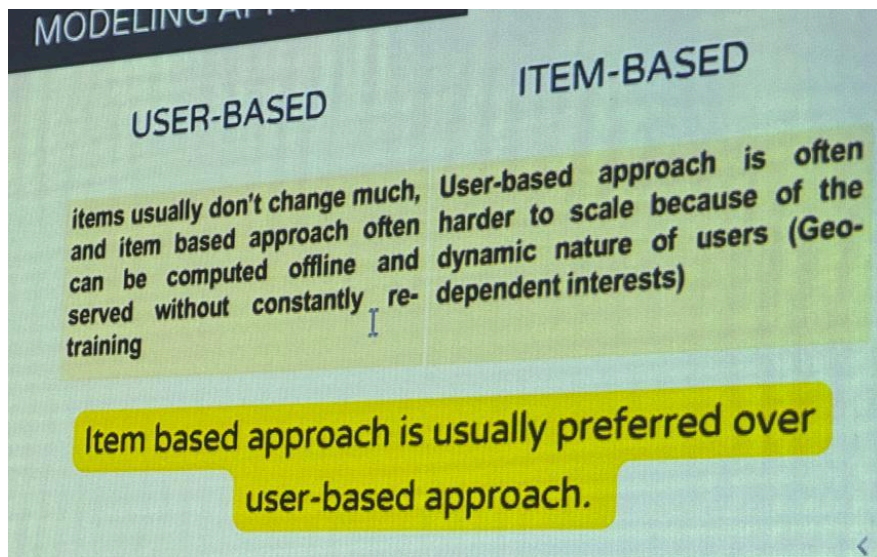
Scalability - big data (a large amount of items and user amount)

Sparsity - how many of you actually rated a product

3. Discuss the limitations of content-based models. (slide 53)

- Limited in scope, for example, it can only make recommendations that are similar to the original seed.
- For users with thousands of purchases, however, it's impractical to base a query on all the items.
- The algorithm must use a subset or summary of the data, reducing quality.
- In all cases, recommendation quality is relatively poor.
- Recommendations should help a customer find and discover new, relevant, and interesting items.

4. There are two types of collaborative filtering models as follows. Differentiate these two types with the aid of suitable examples. (slide 61)
- a) user-user
 - b) item-item



a) user-user

- "People like you also like this."
- Finds similar users, good when fewer users than items.
- Drawback: New user = expensive updates.

b) item-item

- "If you like this, you may also like that."
- Find similar items, good when fewer items than users.
- Drawback: Hard to update when new items appear.

Tutorial 12

Part I: Probability Theory

1. **FOUR (4)** of the rules in a rule-based system are given as follows:

R1:	If there was drizzle last night, then there is 60% chance that the grass in the backyard is wet the next morning.
R2:	If there was rain last night, then there is 80% chance that the grass in the backyard is wet the next morning.
R3:	Drizzle occurs in 160 days in a year.
R4:	Rain occurs in 120 days in a year.

Assume that the drizzle and rain occur exclusively, and there are 365 days in a year.

- Specify the above rules in probability assertion. ()
- If you see the grass in the backyard is wet in the morning, by using Bayes Theorem, what is the event that most probably occurred last night? ()
- After a conclusion is drawn from Question 1(b), your grandfather asserts that sometimes your neighbour would spray water in the backyard at night. ()
 - Do you think this assertion would affect the conditional probability?
 - Do you think this assertion would affect your initial conclusion as well?

Explain your answers.

- a) Specify the above rules in probability assertion. ()

$$P(\text{wet E} \mid \text{drizzle H}) = 0.6$$

$$P(\text{wet E} \mid \text{rain H}) = 0.8$$

$$P(\text{drizzle}) = 160/365 = 0.438$$

$$P(\text{rain}) = 120/365 = 0.329$$

$$P(H | E) = \frac{P(E | H) * P(H)}{P(E)}$$

b)

b) If you see the grass in the backyard is wet in the morning, by using Bayes Theorem, what is the event that most probably occurred last night?

Single evidence with multiple hypotheses

$$P(\text{drizzle} | \text{wet } E) = \frac{P(\text{wet} | \text{drizzle}) * P(\text{drizzle})}{P(\text{wet} | \text{drizzle}) * P(\text{drizzle}) + P(\text{wet} | \text{rain}) * P(\text{rain})}$$

$$P(\text{drizzle} | \text{wet } E) = \frac{0.6 * 0.438}{0.6 * 0.438 + 0.8 * 0.329} = \frac{0.2628}{0.526}$$

$$= 0.4996 \approx 0.5$$

$$P(\text{rain} | \text{wet } E) = \frac{P(\text{wet} | \text{rain}) * P(\text{rain})}{P(\text{wet} | \text{drizzle}) * P(\text{drizzle}) + P(\text{wet} | \text{rain}) * P(\text{rain})}$$

$$P(\text{rain} | \text{wet } E) = \frac{0.8 * 0.329}{0.6 * 0.438 + 0.8 * 0.329} = \frac{0.2632}{0.526}$$

$$= 0.50$$

It can be either rain or drizzle. No significant difference between the two events

- c) i. Do you think this assertion would affect the conditional probability?
- The conditional probability P (wet / spray water) will be added.
 - Hespray water will be added as the additional hypothesis in the problem
- ii. Do you think this assertion would affect your initial conclusion as well?
- The conclusion (posterior probability) will be affected by the frequency of the neighbour spraying water at the backyard at night.
 - If the neighbour rarely sprays water, P(spray) is very low, so it barely affects the other hypotheses.
 - If the neighbour sprays water often, P(spray) is very high, so it strongly reduces the other hypotheses.

2. Part of the rules in a liver disease diagnosis system is shown below. ()

R1:	IF	patient has loss of appetite
	THEN	patient has liver disease
R2:	IF	patient is a alcoholic
	AND	patient is a smoker
	THEN	patient has liver disease

Considering only the following information is given to you.

- $P(\text{patient has loss appetite} \mid \text{patient has liver disease}) = 0.8$
- $P(\text{patient has liver disease}) = 0.00013$

Assume that a patient rated that he has loss appetite 70% of the time, demonstrate how do you predict $P(\text{patient has liver disease} \mid \text{patient has loss appetite})$ using Naïve Bayes Algorithm.

$$P(H \mid E) = \frac{P(E \mid H) * P(H)}{P(E)}$$

$P(\text{patient has liver disease} \mid \text{patient has loss appetite})$

$[P(\text{patient has loss disease} \mid \text{patient has liver appetite}) * P(\text{patient has liver disease})]$

$/ P(\text{patient has loss appetite})$

$= 0.8 * 0.00013 / 0.7$

$= 0.0001486$

3. A police officer sees you run a red light late one Saturday night. Before pulling you over, the officer tries to guess what caused you to run the red light and comes up with the following possibilities:
- You are drunk
 - You are in a hurry
 - You are not wearing your glasses
 - You felt you could do it and get away with it

The policeman assigns the following evidential probabilities to these hypotheses given that the incident happened late on a Saturday night

- $P(\text{run light late Sat. night} \mid \text{drunk driver}) = 0.45$
- $P(\text{run light late Sat. night} \mid \text{driver in a hurry}) = 0.60$
- $P(\text{run light late Sat. night} \mid \text{driver didn't see light}) = 0.15$
- $P(\text{run light late Sat. night} \mid \text{didn't see cop/thinks can get away with it}) = 0.05$

He further assigns the following prior probabilities:

- $P(\text{drunk driver}) = 0.10$
- $P(\text{hurried driver}) = 0.33$
- $P(\text{driver not wearing glasses}) = 0.10$
- $P(\text{driver feels he can get away with things}) = 0.25$

Compute the conditional probabilities for the four possibilities. What decision is the police officer most likely to make? ()

H1 = drunk driver

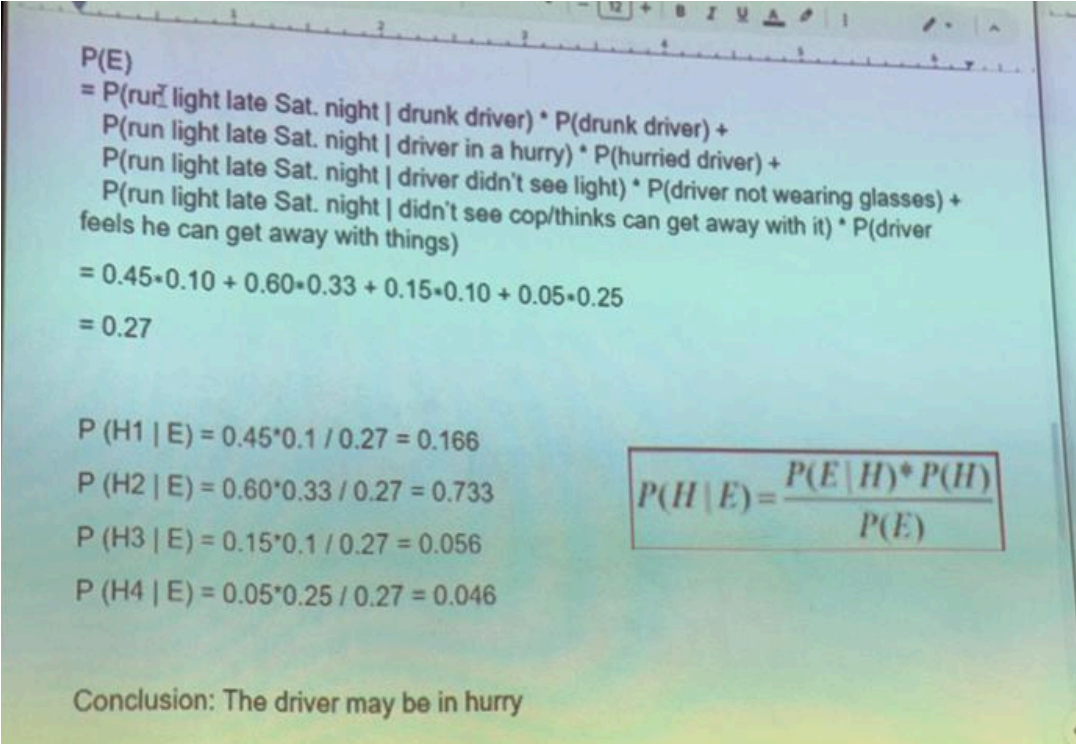
H2 = hurried driver

H3 = driver not wearing glasses

H4 = driver feels he can get away with things

E = run light late Sat. Night

$$P(H \mid E) = \frac{P(E \mid H) * P(H)}{P(E)}$$



$P(E)$
 $= P(\text{run light late Sat. night} \mid \text{drunk driver}) * P(\text{drunk driver}) +$
 $P(\text{run light late Sat. night} \mid \text{driver in a hurry}) * P(\text{hurried driver}) +$
 $P(\text{run light late Sat. night} \mid \text{driver didn't see light}) * P(\text{driver not wearing glasses}) +$
 $P(\text{run light late Sat. night} \mid \text{didn't see cop/thinks can get away with it}) * P(\text{driver feels he can get away with things})$
 $= 0.45 * 0.10 + 0.60 * 0.33 + 0.15 * 0.10 + 0.05 * 0.25$
 $= 0.27$

$P(H1 \mid E) = 0.45 * 0.1 / 0.27 = 0.166$
 $P(H2 \mid E) = 0.60 * 0.33 / 0.27 = 0.733$
 $P(H3 \mid E) = 0.15 * 0.1 / 0.27 = 0.056$
 $P(H4 \mid E) = 0.05 * 0.25 / 0.27 = 0.046$

Conclusion: The driver may be in hurry

Part II: Certainty Factor

4. You receive a phone call from your classmate, James, informing you that your name is in the pass list. Preparing to tell your parents the good news, you recall that James is a liar who loves to play jokes on others. Thus, you decide to call another classmate, Sean. Although Sean is known for being occasionally careless, you believe that Sean is far more reliable. ()

R1: If James' call is true (1.0) , then there is a 0.5 certainty that your name is in the pass list.
R2: If Sean's call is true (1.0) and Sean is not careless is true (0.5) , then there is a 0.9 certainty that your name is in the pass list.
R3: If your name is in the pass list, then there is a 0.99 certainty that you pass the examination.

Conditions:

The certainty of James' call is 1.0.

The certainty of Sean's call is 1.0.

The certainty of Sean is not careless is 0.5

Based on the information provided, would you pass the examination?

CF (1) = CF (names in the pass list – James' call true)
 $= 0.5 * 1$
 $= 0.5$

CF (2) = CF (name in the pass list – Sean's call true AND Sean not careless)
 $= \text{MIN} [1.0, 0.5] * 0.9$
 $= 0.5 * 0.9$
 $= 0.45$

Rules:
1. $CF(A \text{ and } B) = \text{min}(CF(A), CF(B))$
2. $CF(A \text{ or } B) = \text{max}(CF(A), CF(B))$

CF (overall - name in the pass list)
 $= CF1 + CF2 - CF1 * CF2$
 $= 0.5 + 0.45 - 0.5 * 0.45$
 $= 0.725$

CF (Pass)
 $= 0.99 * 0.725$
 $= 0.72$

Conclusion: almost certain I would pass the exam

Formulas for combining CFs:
 $CF1 + CF2 - CF1 * CF2$ if both are positive,
 $CF1 + CF2 + CF1 * CF2$ if both are negative,
 $\frac{CF1 + CF2}{1 - \text{min}(|CF1|, |CF2|)}$ otherwise

Term	Certainty Factor
Definitely not	-1.0
Almost certainly not	-0.8
Probably not	-0.6
Maybe not	-0.4
Unknown	-0.2 to +0.2
Maybe	+0.4
Probably	+0.6
Almost certainly	+0.8
Definitely	+1.0

5. Assume that a simple inference system is built using certainty factors. Demonstrate how to compute the certainty factor that a patient has liver disease based on the following rules. ()

R1:	IF	patient has loss of appetite [CF 0.7]
	THEN	patient has liver disease [CF 0.5]
R2:	IF	patient is a alcoholic [CF 0.8]
	AND	patient is a smoker [CF 1.0]
	THEN	patient has liver disease [CF 0.9]

Combining Certainty Factors:

$$\begin{array}{ll} CF1 + CF2 - CF1*CF2 & \text{if both are positive,} \\ CF1 + CF2 + CF1*CF2 & \text{if both are negative,} \\ \frac{CF1 + CF2}{1 - \min(|CF1|, |CF2|)} & \text{otherwise} \end{array}$$

The image shows a handwritten calculation on a light blue background. On the left, the certainty factors for two rules are calculated. Rule 1 (R1) has a CF of 0.7 for 'loss of appetite' and a CF of 0.5 for 'liver disease', resulting in a combined CF of 0.35. Rule 2 (R2) has a CF of 0.8 for 'alcoholic' and a CF of 1.0 for 'smoker', resulting in a combined CF of 0.72. These two results are then combined using the formula for positive CFs: $CF1 + CF2 - CF1 * CF2$, yielding a final CF of 0.818. On the right, there are two yellow sticky notes. The first note lists two rules: 1. $CF(A \text{ and } B) = \min(CF(A), CF(B))$ and 2. $CF(A \text{ or } B) = \max(CF(A), CF(B))$. The second note is a larger version of the CF combination formula shown in the text above, with the same three cases: positive, negative, and otherwise.

CF (R1)
= 0.7 * 0.5
= 0.35

CF (R2)
= MIN [0.8, 1.0] * 0.9
= 0.8 * 0.9
= 0.72

CF (R1, R2)
= CF1 + CF2 - CF1 * CF2
= 0.35 + 0.72 - 0.35 * 0.72
= 0.818

Rules:
1. $CF(A \text{ and } B) = \min(CF(A), CF(B))$
2. $CF(A \text{ or } B) = \max(CF(A), CF(B))$

$CF1 + CF2 - CF1*CF2$ if both are positive,
 $CF1 + CF2 + CF1*CF2$ if both are negative,
 $\frac{CF1 + CF2}{1 - \min(|CF1|, |CF2|)}$ otherwise

6. The Ebola outbreak has raised concerns of international spread in 2014, similar to the SARS epidemic in more than 10 years ago. Assume that you are assisting a medical centre to develop an online expert system that consults people on recognizing the symptoms of these diseases. Part of the rules of the system is as follows: ()

R1:	IF	body temperature exceeds 37 °C
	AND	headache presents AND muscle aches presents
	THEN	fever presents (CF 0.8)
R2:	IF	body temperature exceeds 37 °C [1.0]
	THEN	fever presents (CF 0.2)
R3:	IF	body temperature exceeds 38 °C [1.0]
	THEN	fever presents (CF 1.0)
R4:	IF	cough presents AND fever presents [1.0]
	THEN	symptom of SARS (CF 0.5)
R5:	IF	cough presents AND cough starts 2-3 days after other symptoms
	AND	fever presents
	THEN	symptom of SARS (CF 0.6)
R6:	IF	IF (cough presents OR nausea presents OR stomach pain presents)
	AND	fever presents
	THEN	symptom of Ebola (CF 0.5)
R7:	IF	IF cough presents AND nausea presents AND stomach pain presents
	AND	fever presents
	THEN	symptom of Ebola (CF 0.8)

John has recently returned from overseas and he started to show the following symptoms. Based on R1 to R7, demonstrate the inference steps and conclude what disease that John could be infected.

Body temperature exceeds 37 °C	CF = 1.0
Headache presents	CF = 1.0
Muscle aches presents	CF = 1.0
Body temperature exceeds 38 °C	CF = 1.0
Cough presents	CF = 1.0
Cough starts 2-3 days after other symptoms	CF = 0.8
Nausea presents	CF = 0
Stomach pain presents	CF = 0.9

Remark: To calculate a combined certainty factor, use the following equation:

$$\begin{array}{ll} CF1 + CF2 - CF1 * CF2 & \text{if both are positive,} \\ CF1 + CF2 + CF1 * CF2 & \text{if both are negative,} \\ \frac{CF1 + CF2}{1 - \min(|CF1|, |CF2|)} & \text{otherwise} \end{array}$$

$$\begin{aligned} \text{R1: CF (fever)} \\ &= \text{MIN} [1.0, 1.0, 1.0] * 0.8 \\ &= 1.0 * 0.8 \\ &= 0.8 \end{aligned}$$

$$\begin{aligned} \text{R2: CF (fever)} \\ &= 1.0 * 0.2 \\ &= 0.2 \end{aligned}$$

$$\begin{aligned} \text{R3: CF (fever)} \\ &= 1.0 * 1.0 \\ &= 1.0 \end{aligned}$$

$$\begin{aligned} \text{R1, R2: CF (overall fever)} \\ &= \text{CF1} + \text{CF2} - \text{CF1} * \text{CF2} \\ &= 0.8 + 0.2 - 0.8 * 0.2 \\ &= 0.84 \end{aligned}$$

$$\begin{aligned} \text{R1, R2, R3: CF (overall fever)} \\ &= \text{CF1} + \text{CF2} - \text{CF1} * \text{CF2} \\ &= 0.84 + 1.0 - 0.84 * 1.0 \\ &= 1 \end{aligned}$$

Rules:

1. $\text{CF}(\text{A and B}) = \text{min}(\text{CF}(\text{A}), \text{CF}(\text{B}))$
2. $\text{CF}(\text{A or B}) = \text{max}(\text{CF}(\text{A}), \text{CF}(\text{B}))$

$\text{CF1} + \text{CF2} - \text{CF1} * \text{CF2}$	if both are positive,
$\text{CF1} + \text{CF2} + \text{CF1} * \text{CF2}$	if both are negative,
$\frac{\text{CF1} + \text{CF2}}{1 - \text{min}(\text{CF1} , \text{CF2})}$	otherwise

$$\begin{aligned} \text{R4: CF (SARS)} \\ &= \text{MIN} [1.0, 1.0] * 0.5 \\ &= 1.0 * 0.5 \\ &= 0.5 \end{aligned}$$

$$\begin{aligned} \text{R5: CF (SARS)} \\ &= \text{MIN} [1.0, 0.8, 1.0] * 0.6 \\ &= 0.8 * 0.6 \\ &= 0.48 \end{aligned}$$

$$\begin{aligned} \text{R4, R5: CF (overall SARS)} \\ &= \text{CF1} + \text{CF2} - \text{CF1} * \text{CF2} \\ &= 0.5 + 0.48 - 0.5 * 0.48 \\ &= 0.74 \end{aligned}$$

$$\begin{aligned} \text{R6: CF (Ebola)} \\ &= \text{MIN} [(\text{MAX} [1.0, 0, 0.9]), 1.0] * 0.5 \\ &= \text{MIN} [1.0, 1.0] * 0.5 \\ &= 1.0 * 0.5 \\ &= 0.5 \end{aligned}$$

$$\begin{aligned} \text{R7: CF (Ebola)} \\ &= \text{MIN} [1.0, 0, 0.9, 1.0] * 0.8 \\ &= 0 * 0.8 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{R6, R7: CF (overall Ebola)} \\ &= \text{CF1} + \text{CF2} - \text{CF1} * \text{CF2} \\ &= 0.5 + 0 - 0.5 * 0 \\ &= 0.5 \end{aligned}$$

Conclusion: John could be infected by SARS (CF 0.74) than Ebola (CF 0.5)

7. Given the report of an observation at 7am as below, demonstrate the inference and conclude the weather of tomorrow based on certainty factors. ()

Place of observation: Miami, Florida (tropical climate) (CF=1)

Observation data:

- Cloud height > 18,000 feet (CF = 0.8)
- Clouds are small, rounded puffs, appeared in long rows. (CF=0.8)
- Cloud patterns are sheet-like. (CF = 0.2)
- Color of cloud is white. (CF=0.7)
- Sun can shine through. (CF=0.9)
- Humidity: warm and sticky (CF=0.7)

Remark: The certain factor (CF) for other unknown data is 0.

IF Cloud_Height > 18,000 feet [0.8] [P1.1]
THEN Cloud_Group is Cirrus [H1] (CF= 0.8)

IF Cloud_Height is between 6,500 feet to 18,000 feet [P2.1]
THEN Cloud_Group is Alto [H2] (CF=0.8)

IF Cloud_Group is Cirrus [P3.1]
AND Cloud_Pattern is sheet-like [P3.2]
AND Cloud_Color is white [P3.3]
AND Moon or Sun can shine through [P3.4]
THEN Cloud_Type is Cirrostratus [H3] (CF=0.8)

IF Cloud_Group is Cirrus [P4.1]
AND Cloud_Pattern is small, rounded puffs with long rows [P4.2]
AND Cloud_Color is white or gray [P4.3]
THEN Cloud_Type is Cirrocumulus [H4] (CF=0.8)

IF Cloud_Group is Alto [P5.1]
AND Cloud_Color is gray or blue gray [P5.2]
AND Moon or Sun can shine through [P5.3]
THEN Cloud_Type is Altostratus [H5] (CF=0.7)

IF Cloud_Group is Alto [P6.1]
AND Cloud_Color is gray [P6.2]
THEN Cloud_Type is Altocumulus [H6] (CF=0.8)

IF Cloud_Type is Cirrostratus [P7.1]
THEN Rain or snow storm occurs within 12-24 hours [H7] (CF=0.7)

IF Cloud_Type is Cirrocumulus [P8.1]
AND Climate_Zone is NOT of Tropical [P8.2]
THEN Weather is fair but cold [H8] (CF=0.7)

IF Cloud_Type is Cirrocumulus [P9.1]
AND Climate_Zone is of Tropical [P9.2]
THEN Hurricane is approaching [H9] (CF=0.6)

IF Cloud_Type is Altostratus [P10.1]
THEN Storms with continuous rain or snow is reaching [H10] (CF=0.7)

IF Cloud_Type is Altocumulus [P11.1]
AND Morning is warm and sticky [P11.2]
THEN Thunderstorms by late afternoon [H11] (CF=0.7)

Part of the rules in the expert system (Reference: boatsafe.com)

H1: CF (Cirrus)
= $0.8 * 0.8$
= 0.64

H2: CF (Alto)
= 0

H3: CF (Cirrostratus)
= $\text{MIN} [0.64, 0.2, 0.7, 0.9] * 0.8$
= $0.2 * 0.8$
= 0.16

H4: CF (Cirrocumulus)
= $\text{MIN} [0.64, 0.8, 0.7] * 0.8$
= $0.64 * 0.8$
= 0.512

H5: CF (Altostratus)
= $\text{MIN} [0, 0, 0.9] * 0.7$
= $0 * 0.7$
= 0

H6: CF (Alto cumulus)
= $\text{MIN} [0, 0] * 0.8$
= $0 * 0.8$
= 0

Rules:

1. $\text{CF}(A \text{ and } B) = \text{min}(\text{CF}(A), \text{CF}(B))$
2. $\text{CF}(A \text{ or } B) = \text{max}(\text{CF}(A), \text{CF}(B))$

$\text{CF1} + \text{CF2} - \text{CF1} * \text{CF2}$	if both are positive,
$\text{CF1} + \text{CF2} + \text{CF1} * \text{CF2}$	if both are negative,
$\frac{\text{CF1} + \text{CF2}}{1 - \text{min}(\text{CF1} , \text{CF2})}$	otherwise

H7: CF (Rain or snow storm occurs within 12-24 hours)
= $0.16 * 0.7$
= 0.112

H8: CF (Weather is fair but cold)
= $\text{MIN} [0.512, -1] * 0.7$
= $-1 * 0.7$
= -0.7

H9: CF (Hurricane is approaching)
= $\text{MAX} [0.512, 0] * 0.6$
= $0.512 * 0.6$
= 0.3072

H10: CF (Storms with continuous rain or snow is reaching)
= $0 * 0.7$
= 0

H11: CF (Thunderstorms by late afternoon)
= $\text{MIN} [0, 0.7] * 0.7$
= $0 * 0.7$
= 0

Conclusion: hurricane may be approaching (CF 0.3072)

Tutorial 13

1. Following shows a sample crisp rule. ()

IF temperature > 37.5
THEN fever = TRUE

- Explain the difference between crisp rule and fuzzy rule.
- Suggest how do you convert the crisp rule above into a fuzzy rule.
- Design the appropriate fuzzy sets to represent the input and output variables respectively.

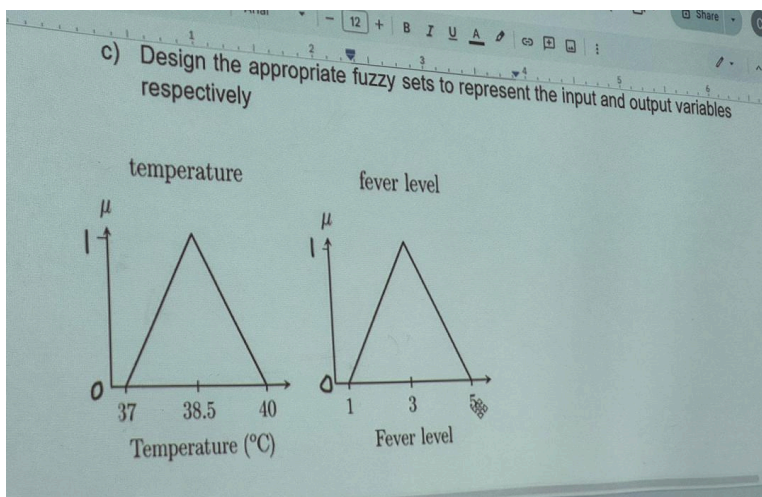
IF temperature > 37.5
THEN fever = TRUE

a) Explain the difference between crisp rule and fuzzy rule.

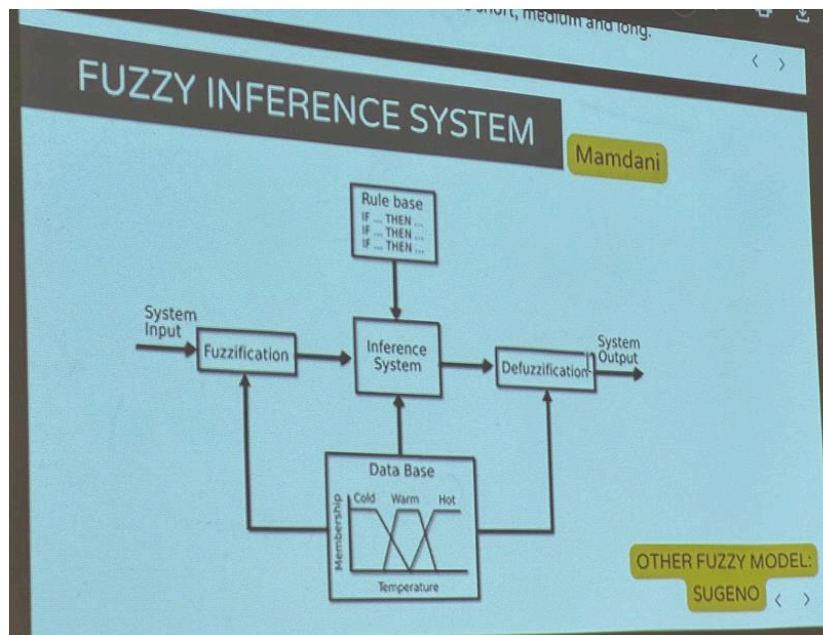
Crisp rules use binary logic, where "x is A" and "y is B" are only true or false. Fuzzy rules use membership functions with values between 0 and 1, so "x is A" and "y is B" can be true to a degree.

b) Suggest how do you convert the crisp rule above into a fuzzy rule.

IF temperature is high
THEN fever is severe



2. Illustrate the general structure of Fuzzy Inference System and explain each of the components. ()



Fuzzification: Converts crisp inputs into fuzzy values using membership functions (e.g., assigns degrees of membership to each fuzzy set).

Membership Function: A curve showing how each input maps to a membership value between 0 and 1.

Rule Base: A set of fuzzy IF-THEN rules (e.g., IF x is A THEN y is B) using linguistic variables and values.

Inference System: Applies fuzzy rules to the fuzzified inputs to produce fuzzy outputs.

Defuzzification: Converts fuzzy outputs back into a single crisp value (e.g., using the Centre of Gravity method).

3. Assume that you are designing an automated air conditioning system using Fuzzy Logic. ()
- a) Suggest an uncertainty in the above scenario with explanation.
 - b) Generate the **TWO (2)** input and **TWO (2)** output fuzzy membership functions that are able to model the air conditioning system.
 - c) Suggest **TWO (2)** fuzzy rules that associate to the Fuzzy membership functions generated in Question 3(b).

a) Temperature can be seen at different levels of coldness. For example, is 21 °C considered cold?

b) Input: Temperature of the room (cold, cool, comfortable, warm, hot), humidity (dry, moderate, humid)

Output: Cooling power (low, medium, high), fan speed (low, medium, high)

c) IF Temperature is Hot OR Humidity is Humid

THEN Cooling Power is High AND Fan Speed is High.

IF Temperature is Comfortable AND Humidity is Moderate

THEN Cooling Power is Medium AND Fan Speed is Medium.

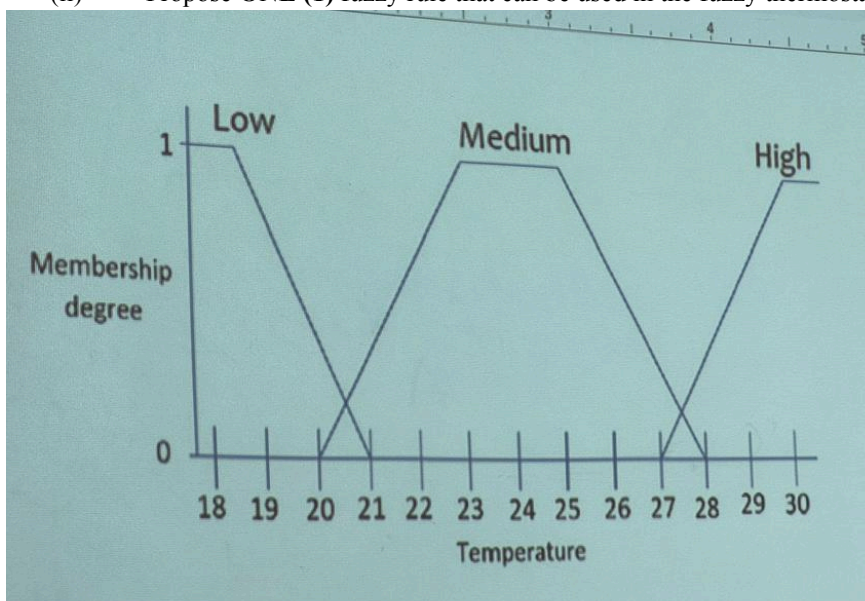
4. Suppose you are designing a fuzzy thermostat with the following membership function of the temperature. ()

Low(x) = $\begin{cases} 0, & \text{if temp (x) } \geq 21, \\ f(\text{temp(x)}), & \text{if } 19 < \text{temp(x)} < 21, \\ 1, & \text{if temp(x) } \leq 19 \end{cases}$

Medium(x) = $\begin{cases} 0, & \text{if temp (x) } \leq 20 \text{ or temp (x) } \geq 28, \\ f(\text{temp(x)}), & \text{if } 20 < \text{temp(x)} < 23 \text{ or } 25 < \text{temp(x)} < 28 \\ 1, & \text{if } 23 \leq \text{temp(x)} \leq 25 \end{cases}$

High(x) = $\begin{cases} 0, & \text{if temp (x) } \leq 27, \\ f(\text{temp(x)}), & \text{if } 27 < \text{temp(x)} < 30 \\ 1, & \text{if temp(x) } \geq 30 \end{cases}$

- (i) Illustrate the graph with labels to represent the above membership function.
(ii) Propose **ONE (1)** fuzzy rule that can be used in the fuzzy thermostat.



- ii) If the room temperature is warm then the fan speed is high